Coordinated Great Lakes

Regulation and Routing Model

Draft User's Manual

March 20

1 BACKGROUND

1.1 History

Throughout the government agencies overseeing hydraulic and hydrologic issues within the Great Lakes basin, there was known to exist several regulation and routing models. These models were written in various programming languages, by many different people, using many different styles. Due, in part, to personnel changes within these various agencies, oftentimes these models lacked the documentation needed for new users to easily understand and use the models.

During the 1990s, the binational Coordinating Committee on Basic Great Lakes Hydraulic and Hydrologic Data (CCBGLHHD) recognized the need to develop and maintain a comprehensive model that would permit coordination and collaboration among the government agencies overseeing Great Lakes issues. They also recognized that outside researchers and the public should also have access to this model.

One of the main goals of the CGLRRM was to replace many existing models with one consistent, coordinated effort. However, given that the CGLRRM would compute period average levels and outflows when given supplies and regulation plan parameters as input, there would be limitations. Computations would not reflect short-term hydrodynamic effects such as wind setup, ice jams, etc. Nonetheless, the model would provide reliable averages for periods of approximately a week or more.

1.2 Participating Agencies

An ad-hoc working group formed to oversee model development reports to the binational Coordinating Committee on Basic Great Lakes Hydraulic and Hydrologic Data. U.S. and Canadian government agencies represented by working group members currently include:

Environment Canada – Great Lakes-St. Lawrence Regulation Office, Cornwall, ON
Great Lakes Environmental Research Laboratory – National Oceanic and Atmospheric Administration, Ann Arbor, MI
U.S. Army Corps of Engineers – Buffalo District, Buffalo, NY
U.S. Army Corps of Engineers – Detroit District, Detroit, MI
U.S. Army Corps of Engineers – Hydrologic Engineering Center, Davis, CA

The Detroit District of the USACE currently acts as model custodian.

1.3 Model Predecessors

Each of the government agencies listed above have developed multiple models for use during their regulatory, research, and study operations. Hydraulic models range from simple stage-fall-discharge relationships to 2-D models of the rivers. Hydrologic models range from small watershed simulation models to GLERL's Advanced Hydrologic Prediction System (AHPS), a comprehensive model used to predict supplies on each of the Great Lakes from meteorological conditions from over a hundred subbasins and each lake. Many of these models are used during operations on a regular basis, though many were designed for particular purposes during applied studies.

1.4 Model Development

An ad-hoc working group comprised of modelers from each of the government agencies was formed to oversee model development. All development issues have been coordinated by the working group. Mr. Matt McPherson, a USACE engineer (HEC; formerly with Detroit District), has been the main programmer who structured the model and developed the section from Lake Superior to Lake Erie. Mr. Paul Yu, a USACE (Buffalo District) engineer, programmed the Lake Ontario/St. Lawrence River sections of the model.

The first phase of development incorporated GLERL's middle lakes routing models called MIDLAKES. This model calculates the levels in the Lakes Michigan-Huron, St. Clair, and Erie portion of the system along with the flows in the St. Clair, Detroit, and Niagara Rivers. The quarter-monthly version is described in GLERL's technical paper ERL-GLERL 109. The MIDLAKES model incorporated into the CGLRRM includes flexible input and a matrix algebra solution to the simultaneous equations used for midlakes routing, daily routing, and generalized outflow equation formulas.

In the second phase, USACE Detroit District integrated the middle lakes portion with code for Lake Superior regulation Plan 1977-A into a single modular architecture.

The third phase was the addition Lake Ontario/St. Lawrence River Regulation Plan 1958-D, with the work done by USACE Buffalo District and coordinated with Environment Canada. The work at the completion of Phase III consisted of Fortran 77 programs with ASCII inputs and outputs, suitable for the expert users within the agencies.

The fourth phase will provide a user-friendly graphical user interface (GUI) to simplify operation and output analysis by more casual users. The basic scheme is to develop front- and back-end interfaces to create the ASCII input files, run the model, and process the ASCII output files. This design separates the core model from the GUI. GUI design issues are currently being addressed, and a basic conceptual design should be available by early Fall 2001.

2 SCOPE OF THE MODEL

In general, given the parameters describing the hydraulic characteristics of the system, and hydrologic input data, the model will simulate Lake Superior regulation, route flows through Lakes Michigan-Huron, St. Clair, and Erie, and simulate Lake Ontario regulation.

The CGLRRM replaces many existing models with one consistent, coordinated package. However, given that the CGLRRM computes period average levels and outflows when given supplies and regulation plan parameters as input, there are limitations. Computations do not reflect short-term hydrodynamic effects such as wind setup, ice jams, etc. Nonetheless, the model provides reliable averages for periods of approximately a week or more.

2.1 Model Components

The model consists of three main modules, delineated by geographical location:

1. The Lake Superior module permits routing of basin supplies through the uppermost lake, with outflow conditions defined by either the preproject (1887) equation or Plan 1977-A regulation conditions. Channel routing through the St. Mary's River (downstream of Lake Superior) is included in this module.
2. The middle lakes module permits routing of supplies through Lakes Michigan and Huron (hydraulically one lake), St. Clair, and Erie, as well as the connecting channels (i.e., the St. Clair, Detroit, and Niagara Rivers). Routing can be performed using either the iterative solution of solving nine simultaneous equations, or GLERL's MIDLAKES matrix routine. Outflow from the Niagara River can be controlled using several different equations (see Section 4.3.3 for more information).
3. The Lake Ontario/St. Lawrence River module permits routing of supplies through the downstream end of the Great Lakes system. Niagara Falls acts as a convenient, natural hydraulic break between this module and the other two. Again, either the preproject relations, or Lake Ontario regulation Plan 1958-D may define outflow conditions.

Though the model is broken into the above modules (as well as other smaller modules), they are each sequentially linked together to permit forecasting and simulations throughout the entire system, from Lake Superior to Montreal (eventually Trois-Rivieres), Quebec.

Future versions of this manual will include a flowchart to delineate the various modules and other model components.

2.2 Practical Limitations

It is important for users to understand the main purpose of the CGLRRM. That is, given the parameters describing the hydraulic characteristics of the system, and the user-defined hydrologic input data, the model simulates Lake Superior regulation, routes flows through the unregulated outlets of Lakes Michigan-Huron, St. Clair, and Erie, and simulates Lake Ontario regulation.

Since the CGLRRM is strictly meant to be a routing and regulation model, there is no ability to consider short-term hydrodynamic anomalies such as wind setup, ice jams, peaking and ponding, etc. The CGLRRM computes period average levels and outflows from given supplies and regulation plan parameters (i.e., there are no hydrodynamic computations; just water balance calculations). Owing to these common and oftentimes unexpected or unknown secondary hydrodynamic driving forces, users should only consider averages for periods of approximately a week or more as reliable results.

In addition to the current regulatory plans for Lake Superior (i.e., Plan 1977-A) and Lake Ontario (i.e., Plan 1958-D) that are included, an option is available to have outflows from these two lakes determined by the unregulated outlet (i.e., preproject) conditions. As such, users should be aware that use of supply sequences outside of the ranges used in development of either plan (e.g., extreme climate change scenarios beyond those found in the relevant preproject periods of record used) may result in impractical output values. For instance, a very lengthy low supply sequence routed through Lake Ontario could result in the lake level falling below the outlet sill elevation ("draining of the lake") since Plan 1958-D has a set of minimum outflow limits.

Simulations are also limited by the model assumptions, which impose constant physical outflow relationships throughout the entire simulation period. For example, though it may be desired to replicate monthly mean water levels on Lake St. Clair for the entire 20th century, this simulation must be performed with one set of channel capacities for the St. Clair and Detroit Rivers, although the channels were enlarged several times since the early 1900s.

Stage-fall-flow relations developed for the connecting channels were developed under certain ranges of water levels and outflows. Use of these relations for conditions beyond which they were developed will result in impractical outflow estimates in many cases. For instance, imposing extremely high Detroit River water level input values (that would equate to overbank flooding) would result in gross errors in Lake Erie inflow estimates.

3 GETTING STARTED

3.1 System Requirements

A Windows-based personal computer is recommended. Users may opt to employ the model on most other systems, provided they have a Fortran compiler and simple shell operations in order to install the package. Special installation instructions are given for non-Windows users at the end of Section 3.2.

Note that a graphical user interface has yet to be developed to simplify model use. Currently, it is still necessary to traverse directories and edit ASCII files to run the model and view results.

The only actual system requirement is that a minimum of approximately 15 MB of hard disk drive space should be available.

3.2 Installation

Windows users

1. Enter Windows Explorer and create a new temporary folder.
2. Download the self-extracting archive file http://www.lre.usace.army.mil/cglrrm/cglrrm-i.exe into this folder.
3. Run this executable file to extract the setup files.
4. Run "setup.exe" and follow the prompts.

Non-Windows Users

Users must have a Fortran compiler and knowledge of simple shell operations in order to install the package. The following instructions pertain to Unix systems, but apply to all systems with minimal adaptation.

1. Create a directory (e.g., "/wcds/cglrrm").
2. Download the distribution package (cglrrm-i.tar.Z) into it.
3. Uncompress the distribution package (enter "compress -d cglrrm-i.tar.Z).
4. Untar the distribution package (enter "tar-xvf cglrrm-i.tar). This will create subdirectories: "src", "bin", "data", "test", and "demo".
5. Make "src" the present working directory (pwd; enter "cd src") and edit the file "custom.for". This contains a few operating system- and compiler-dependent settings. Read the commented instructions and make all required changes.
6. Compile and link all the source files into an output executable named "cglrrm" in the "bin" directory. (For a SparcStation compiler, the command is:
   f77 -o .../bin/cglrrm cglrrm.for calcutl.for control.for custom.for init.for midlakes.for miscutl.for ontario.for superior.for tsinput.for)

There should be no errors, but possibly some warnings. The “readme.txt” file in the install directory contains compiling tips.

Distribution Package Contents

The "src" directory contains source code files. Users may benefit from browsing the source files and reading the comments embedded there. Besides notes regarding the coding, they contain explanations of modelling techniques and data treatments.

The "data" directory contains a number of time-series data files: net basin supplies (NBS) for Lakes Michigan-Huron, St. Clair, and Erie; St. Mary's River flows; discharge rates for the Welland Canal and Chicago Diversion; and ice/weed flow retardation amounts for the St. Clair, Detroit, and Niagara Rivers. These files illustrate the monthly, quarter-monthly, weekly, and daily formats for time-series data.

The "op" and "demo" directories are sample locations to make model runs from. The "demo" directory contains two files. "cglrrm.ini" is the master input file, referencing "../bin/system.dat" and "demo.dat". The master input file (always "cglrrm.ini") must be located in the present working directory (pwd) when you execute the model, and lists the files that the run parameters are read from. "demo.dat" is a big parameter file for a run through the upper lakes, demonstrating various capabilities of the model. New users should read it and cross-reference the various entries in Section 4.3. The "op" directory is similar, except that it illustrates the parameters required to make a Lake Superior regulation run for a single month. It holds a much smaller parameter file named “changeme.dat", limited to data that changes from month-to-month. The master input file specifies a third parameter file, ".../bin/operate.dat", that contains the parameters that remain constant for any such run. The "supreg" directory illustrates the parameters required to perform the three traditional Lake Superior regulation runs, analyzing 5%, 50%, and 95% supply scenarios for six months. The first month of the 50% supply scenario yields results identical to those in the "op" directory. Although the regulation decision is based entirely on this one month of results, the other 17 months of forecast results are used for information and reporting.

The Lake Ontario sample dataset is included to demonstrate the I/O as it currently stands. As the Ontario/St. Lawrence module matures it will need substantial attention.

More information about each run is available in readme files inside each directory.

The "bin" directory actually contains ASCII files (not binary). File "messbase.txt" lists hundreds of helpful error and warning messages. "system.dat" is a parameter file containing information specific to each installation, and must be specified at the top of each "cglrrm.ini" file for each run. In this file, a crucial line beginning "Message Database:" specifies the path to the message database (i.e., "messbase.txt"). Usually it will be necessary to change the default ("...\bin\messbase.txt") to the appropriate one (e.g., "c:\cglrrm\bin\messbase.txt"). The Windows install package does this automatically. Also in this directory are some miscellaneous files that may be useful. One is a small DOS shell script named "compare.bat", which can be used after running the sample dataset to check results against the sample output. "grep.com" is a port of the UNIX command, helpful for extracting certain lines from the output files. "list.com" is a fast way to view large files. The model executable ("cglrrmi.exe") also resides in the "bin" directory. It can be run at any time by specifying the complete file specification at the command prompt (e.g., type "C:\CGLRRM\BIN\CGLRRM"). It may be helpful to ensure that this directory is included in the PATH, or create a symbolic link to it in the PATH. In Unix, this is accomplished via an "ln -s source target" command. In Windows, directories are "folders", and symbolic links are called "shortcuts". Run Windows Explorer and click on the directory to be used to create a link. Choose "File", then "New", and "Shortcut" from the menu at the top. Fill in the dialog boxes to locate the CGLRRM executable file and specify a name for the new shortcut.

Finally, move to the directory containing the master input file ("cglrrm.ini") to be used, and run the executable. After model execution, run the comparison script (e.g., "...\bin\compare"). Windows NotePad may be used to display which lines in the output files differ from the "known" outputs sent with the distribution package. If there are differences (other than the date/time stamp) please contact the model custodian.

4 INPUT

4.1 Data File Format

There are three basic types of input files. All lengthy time-series data, such as historical NBS or ice retardation values for a simulation, reside in separate data files. Miscellaneous data (e.g., a lake's surface area or outflow equations for a lake) reside in one or more parameter files. The parameter files contain the names of time-series data files to be read. Finally, a master input file named "cglrrm.ini" contains the names of the parameter files to read. Note that command line arguments (which are not supported in Fortran 77) will eventually be incorporated into the master input file.

The CGLRRM does not support filenames that include spaces.

Time-Series Data File Format

Data files contain the daily, weekly, quarter-monthly, or monthly values needed for a given simulation period. Often they will hold many years of data (only some of which might be used for a particular run). Data files contain only ASCII characters within the range from 32 to 126 (base 10), with a carriage return and/or line feed terminating each record. A value of -9999 must be used to indicate missing data. Regarding net basin supply (NBS), "other" supplies, and "other" diversions (see Section 4.3 for further information); positive values indicate flow into a lake, while negative values indicate flow out of a lake. Consumptive use flows, the Long Lac/Ogoki Diversions, the Chicago Diversion, and Welland Canal flow should always be positive, and this direction information is hardcoded. The program opens, reads, and closes these input data files lake-by-lake in a pre-processing step.

One record near the top of the file must give the units that data is expressed in. The record must begin with the string "UNITS:" followed by one of:

- m3s cubic meters per second (flow)
- 10m3s tens of cubic meters per second
- cfs cubic feet per second
- tcfs thousands of cubic feet per second
- inches inches
- feet feet
- cm centimeters
- mm millimeters
- m meters (distance or elevation)
- m2 square meters (area)
- ft2 square feet
- mi2 square miles
- km2 square kilometers
- m3 cubic meters (volume)
- ft3 cubic feet

The units m3s, m, m2, and m3 are designated as the basic units for flow, distance, area, and volume. All data read in is immediately converted to one of these units, before being used. All data lines must conform to a single, fixed-column format. For monthly data, each record must list the year, followed by twelve monthly values in fields of eight columns each. Quarter-monthly data records must list the year and quarter-month, followed by 12 values (one for each month) applicable to that quarter of the month, also in 8-column fields (see example following). Daily data records must list the year, month, and quarter-month, followed by 7 or 8 values (one for each day) applicable to that quarter of the month, also in 8-column fields. The weekly data format consists of the date (year, month, and day) corresponding to the end of the week, followed by a single 8-column value. For Lake Ontario regulation, weeks always end on a Friday.

```
# LAKE ERIE QUARTER MONTHLY NET BASIN SUPPLIES (10m^3/s)
# Comment field
# 1900      1990
# Fortran format: I4 12F8.0
INTERVAL: Quarter-Monthly
UNITS: 10m3s
# Yr QM      J       F       M       A       M       J       J       A       S
O       N       D
1900 1    -419     102     244     147     113     113      34     -37    -102     -
```

```
71     -65      79
1900 2      96     176     238     147      62      -8     -25      23    -161     -
76    -102     -54
1900 3      99     167     255     153      57     -20      25      74    -227     -
74      57    -116
1900 4     127     246     244     102      54      57      23  -102      48    -142
    23    -88
1901 1      34    -17      28     190      11      40      96    -161     -28    -
210     -85     -91
…
```

Another record near the top of the file must give the time interval of the data (e.g., "Quarter-Monthly" or "Daily"). Only the first letter is significant, and case doesn't matter (must be m, q, w, or d; M, Q, W, or D). The record must begin with the string "INTERVAL:" (see above example).

Any line beginning with a pound character ("#") is ignored as a comment. Use of comment lines is highly encouraged. Data files should provide at least this much documentation near the top:

1) type of data (e.g., "net basin supplies")

2) time period contained in file (e.g., "(January) 1900 to (December) 1990")

3) Fortran format statement (e.g., "I4, 12F8.0"). Here, this example tells the user that each line of data contains a four-character integer, followed by a dozen real numbers; each with eight characters, and no decimal places.

Though not required, it is recommended that these files adhere to a strict naming convention. The first two characters should refer to the location of the data (i.e., "sp" for Lake Superior or for "nr" for Niagara River). The third character indicates the time increment of the data. The allowable values are "m" for monthly, "q" for quarter-monthly, "w" for weekly, "d" for daily, or a hyphen ("-") for mixed time periods or other unusual arrangements. The fourth, fifth, and sixth characters should describe the type of data (e.g., "nbs" for net basin supplies, or “div" for diversion). The seventh and eighth characters in the filename are optional. A three-letter extension is reserved for scenario identification (e.g., coordinated historical Lake Superior monthly net basin supplies might reside in a file named spmnbs.cor; file onwdevec.p98 would contain Lake Ontario weekly deviations from plan (supplied by Environment Canada for use in testing Plan 1998)). A comprehensive list of allowable data file names is given near the end of this section. The program also uses this convention when writing optional output files.

Finally, the end-of-line delimiter for the operating system used must terminate the last line. For Windows users, the end of each line is marked by a carriage return/linefeed pair (ASCII 13, then 10). For Unix users, this is the linefeed character (decimal value 10 in ASCII sequence). Should an error occur regarding reading of data files, load the relevant data file into a text editor, key to the end of the last line, and hit enter. Save the file, then retry running the model.

The CGLRRM does not support filenames that include spaces.

Allowable Filenames for Time-Series Data

A comprehensive list for all possible time-series data (input and output) is included as Appendix F. The first two characters denote the lake or river for which the data is provided. Filenames have an "m", "q", "w", or "d" as the third character (although only the quarter-monthly "q" version of the filename is listed in Appendix F). The seventh and eighth positions, and the three-character extension, are available for optional descriptions.

4.2 Parameter File Format

Parameters include inputs, such as the starting and ending times for a simulation, trigger levels in a regulation plan, and regression constants for stage-discharge equations. The contents of these files will vary somewhat according to need (e.g., a period-of-record simulation may require parameters of no relevance to a 6-month forecast run). The parameters for a run may be contained in a single file, or up to ten separate files. Though guidelines for organizing the parameters are given, the program does not presume any arrangement. The parameters may be placed throughout the file(s) according to user preference (e.g., a forecaster may prefer to organize static inputs for Plan 1977-A, midlakes routing, and Plan 1958-D into separate files, which would require editing very rarely. Parameters such as initial lake levels would belong in a file revised for every run. Users evaluating changes in a regulation plan would prefer much different arrangements). These files have no naming convention, although MS-DOS type 8-character.3-extension names are recommended for portability reasons.

There are five broad categories of input parameters (see Section 4.3 for details). One consists of variables required for routing through the middle lakes, such as whether to use the iterative or matrix solution, or the number of increments to solve for within each period. Parameters associated with Plan1977-A (Lake Superior regulation) or Plan 1958-D (Lake Ontario regulation) comprise two other categories. The fourth group consists of run control parameters, such as output options, starting date, titles, etc. The fifth group is made up of routing parameters whose forms are very similar for each of the lakes (supplies, discharge equation constants). The comprehensive inventory of input parameters is contained in Section 4.3.

The units m3s, m, m2, and m3 are designated as the basic units for flow, distance, area, and volume. All parameters read in will be immediately converted to one of these units, before being used. For a list of acceptable units, see Section 4.1. For parameter files, it is necessary to specify units after many parameter ID codes. Please refer to Section 4.3 for specific examples, as requirements vary from depending on the parameter.

4.3 Description of Input Parameters

The following parameter codes must be the first string on each non-commented line. The codes are not case-sensitive, and only letters and numbers are used to determine a match (dashes, slashes, periods, capitals, and mixed case are ignored). All alphanumeric characters on the line before the first colon (":") must match a parameter code from the following list, or an error message will be issued and the program will terminate. The remainder of the line must contain the values appropriate to that input parameter, in a comma-delimited list.

The order in which parameter records are listed is not significant (if the same parameter is specified more than once, the program stops). The four exceptions are (see Section 3.4 for examples):

1) The "General Verbosity" parameter may be specified as many times as desired in order to debug the parsing of parameters.
2) The "Modeler Name" parameters may be specified multiple times, with the final value overriding any previous input.
3) The area/elevation parameters use one record for each area/elevation pair.
4) The various time-series data parameters may specify repeating values across multiple lines.

In general, default values are not recognized. Therefore, each parameter required for a given application must be explicitly defined for each run.

Time-series data input data (distinguished by ending in the keyword "DATA") may be specified as a filename, or as "repeating data", which consists of an entire year's worth of data values specified on the parameter line (see Section 4.4 for examples). Any repeating values are used for each year in the simulation. The first line of the repeating data will have the parameter ID code, the units of the data (same list of recognized labels as with input file (Section 4.1)), the time interval of the data (i.e., monthly, quarter-monthly), the number of values that follow on this line, and the values. The repeating data can be spread over several lines. Subsequent lines do not require units or time-interval.

A cross-referenced, alphabetic index of all parameter ID codes is given in Appendix A. Note that the item numbers match those listed in the appendix. The first number is the sub-section (e.g., 4.3.1), and the second number is the parameter number within this sub-section.

4.3.1 Run Control Parameters

LAKES TO SOLVE FOR **1-1**

Code for which lake levels to solve for (required). Consists of a group of contiguous letters, after the ":", describing which lakes to model. The letters are "S" for Superior, "M" for Michigan-Huron, "C" for St. Clair, "E" for Erie, "O" for Ontario, and "L" for St. Louis. Currently, if "M", "C", or "E" is specified, the other middle lakes must be specified as well.
Valid codes are:
SMCE- Lakes Superior through Erie.
MCE- Lakes Michigan through Erie.
SMCEO- Lakes Superior through Ontario.
O- Lake Ontario
Example:
LAKES TO SOLVE FOR: SMCE

GENERAL VERBOSITY, TS DATA VERBOSITY, CHECK VERBOSITY, SUP VERBOSITY, MIDLAKE VERBOSITY, ONTVERBOSITY **1-2 to 7**

These parameters control the degree of debugging information written to the screen or the

detailed log file. Values of 1 are most terse, while 9 are most verbose. The value in any given module defaults to General Verbosity, unless overridden by the value appropriate to that module (i.e., "TS Data Verbosity" for processing NBS and other time-series data, "Check Verbosity" for the routine that checks the consistency and completeness of the input, "Sup Verbosity" for the Lake Superior computations, "Midlake Verbosity" for the middle lakes routing, and "Ont Verbosity" for the Lake Ontario/St. Lawrence calculations). If not specified, the default is 5. Note that only the General Verbosity control is available while input parameters are being parsed. This value may be set and reset within the parameter files to provide debugging info as desired, but the LAST value specified will remain in effect for the duration of the computations. Example:
GENERAL VERBOSITY: 9
CHECK VERBOSITY: 6
SUP VERBOSITY: 7
MIDLAKE VERBOSITY: 3
ONT VERBOSITY: 9
TS DATA VERBOSITY: 1

OUTPUT EXTENSION 1-8
Extension to use for output files (optional). A single case-sensitive string of up to six characters, which will be used as the extension on output filenames. Include the leading "." (if applicable). If this parameter is not specified, no extension will be used. Only the first word after the ":" is used; the rest of the line is available for comments. Example:
OUTPUT EXTENSION: .reg (add your comments here!)

MESSAGE DATABASE 1-9
Location of message database (optional). Refers to the filename of the message database, a text file used by the program to provide detailed feedback regarding an error or suspicious circumstance. Though not required, it is useful at times. If undefined, and an error occurs, the program looks for file "messbase.txt" in the present working directory. If not available, a small message appears, and the run continues. Ensure that relative paths originate from the present working directory (pwd) when the program is launched (be careful of case-sensitivity if using non-Windows file systems). The file specification must be less than 80 characters long. This input parameter should be placed at the top of the list, since the program may want to provide information regarding an error experienced while reading the parameters. Recommended practice is to put it in parameter file named "system.dat" or similar, and put the complete file specification at the top of all "CGLRRM.INI" files. No spaces are allowed. Examples:
MESSAGE DATABASE: C:\CGLRRM\BIN\MESSBASE.TXT (Windows)
MESSAGE DATABASE: ...\bin\messbase.txt (DOS-style relative path, pwd=C:\CGLRRM\TEMP)
MESSAGE DATABASE: bin\messbase.txt (similar, but pwd=C:\CGLRRM)
MESSAGE DATABASE: /usr/cglrrm/bin/messbase.txt (Unix absolute path)

OUTPUT DIRECTORY 1-10
Location of output files (optional). Refers to the directory where the output files are to be written. The root name of each output file will be directly appended to this parameter, so a trailing directory separator (i.e., '\' for Windows/DOS or '/' for Unix) must be included. If using relative paths, they must originate at the present working directory (pwd) when the program is

launched. Be careful of case-sensitivity if using non-DOS file systems. The output path must be less than 68 characters long. This parameter does not control the location of log files, which always reside in the present working directory. If not specified, output files are written to the present working directory. Only the first word after the":" is used. Examples:
OUTPUT DIRECTORY: C:\CGLRRM\OUT\ (Windows absolute path)
OUTPUT DIRECTORY: OUT\ (same, if pwd=C:\CGLRRM)
OUTPUT DIRECTORY: …\OUT\ (same, if pwd=C:\CGLRRM\TEMP)
OUTPUT DIRECTORY: /export/home/lreepclw/supreg/output/ (Unix absolute path)
OUTPUT DIRECTORY: …/output/ (same, if pwd=/export/home/lreepclw/supreg)

START DATE 1-11
Start date of simulation (required). Consists of three integers: a four-digit year, an optional one- or two-digit month, and an optional one- or two-digit day of month. If a month is not specified, January is assumed. If a day is not specified (and a weekly, daily, or quarter-monthly output interval is used), the first of the month is assumed. These examples are equivalent:
STARTDATE : 1997, 11, 1
START DATE: 1997, 11

END DATE 1-12
End date of simulation (required). Consists of three integers: a four-digit year, an optional one- or two-digit month, and an optional one- or two-digit day of month. If a month is not specified, December is assumed. If a day is not specified (and a weekly, daily, or quarter-monthly output interval is used), the last day of the month is assumed. These examples are equivalent:
END DATE: 1998
END DATE: 1998, 12, 31
ENDDATE : 1998, 12

TITLE 1, TITLE 2, TITLE 3 1-13 to 15
Descriptive information applicable to the run (optional). Used as part of headings for input descriptions and output summaries. Maximum 79 characters. Examples:
TITLE 1: Historical Simulation of Lake Superior Regulation
TITLE 2: Evaluate Change in Plan 1977-A
TITLE 3 : Study Prompted by Inquiry of 15 Dec 2000

BOM ROUND, LEVEL ROUND, FLOW ROUND 1-16 to 18
Determines number of decimal places left after rounding, expressed as powers of ten (optional). Acceptable entries range from -6 to 3, though -99 indicates no rounding (default). These are global settings and provide some user control regarding nonconformance with existing practice, which rounds intermediate levels to four decimal places, beginning- or end-of-month (BOM and EOM) levels to the centimeter (two decimal places), and flows to integer (meters cubed per second). Operational practices associated with Plan 1977-A depend on using these accepted values (though such practices may be modified in the future). Examples:
BOM ROUND : -2
LEVEL ROUND: -99
FLOW ROUND: 1

MONTH LENGTH **1-17**
Either uniform or actual (optional). Existing practice uses "uniform" month (or quarter-month) length, where February is same number of seconds as January (i.e., the average length of the Gregorian calendar month, with 97 leap days every 400 years. (so month length is: ((400 * 365 + 97) d * (24 * 60 * 60) s/d) / (400 *12) month = 2629746 s/month)). The length of a uniform quarter-month is 1/4 of that. The main impact of this parameter occurs regarding change-in-storage calculations. To replicate existing practice, be sure to use the proper areas (see later), because the month length and water surface areas were traditionally combined in a "storage constant" for each lake. The "actual" month length uses the actual number of seconds in each month (or quarter-month). For January, the number of seconds would be 31 * 24 * 60 *60 = 2678400 s/month. Quarter-months starting/ending dates are1-8, 9-15, 16-23, and 24-last day on month, except February, when the first 3 quarter-months are all seven days long. Applies globally to all routing computations. The first character after the":" must be "A" or "U"; the rest of the line is ignored. Examples:
MONTH LENGTH: Actual
MONTH LENGTH: Uniform

SUP OUTPUT FILES, MHU OUTPUT FILES, STC OUTPUT FILES, ERI OUTPUT FILES, ONT OUTPUT FILES **1-20 to 24**
Indicates which output files are desired for each lake, in addition to the standard results file (optional). Data follows standard formats; the files are named according to the previously defined convention , and written to the output directory . Identified by the first string after the ':'. Only the letter 'D', 'W', 'Q', or 'M' is valid. They must be contiguous, but in any order. The routing interval of the requested output must not be shorter than the computation interval (e.g., can request files of monthly, quarter-monthly, and weekly values from a run using a weekly routing period, but not files of daily values). Examples:
SUP OUTPUT FILES: M (any routing period)
MHU OUTPUT FILES: DMWQ (must be a daily routing)
ERI OUTPUT FILES: MW (must be weekly or daily routing)

4.3.2 Lake Superior Regulation Parameters

STM RIV SOLUTION **2-1**
Determines method to use to calculate Lake Superior outflows (required if routing through Lake Superior). If the first letter after the ':' is an 'O', then an outflow equation (e.g., the preproject 1887 relationship) is used which must be specified using "STM RIV FLOW EQN". If '1977A' is provided, then Plan 1977-A is used. The 1955 Modified Rule of '49 will be an additional option in future versions. Examples:
STM RIV SOLUTION: outflow equation
STM RIV SOLUTION: 1977A

P77A COMPAT MODE **2-2**
Describes which supplies, retardations, and diversions are used to calculate the gated and sidechannel flows for the month (after the Plan 1977-A determination of outflow and number of gates). A setting of “0” indicates that computations should use the values read in, as given in "Sup NBS data", "Mhu NBS data", etc. to make the gate decision for the upcoming month. Though this is the traditional method, the logic is flawed since the user is unaware of the supply

for the upcoming month. To replicate standard six-month forecasts, use 0, with the standard six-month sequence of operational supplies and retardations specified for the various "… NBS Data" data. “-1” applies the "operational" supplies, diversions, and retardations to make the gate decision for the upcoming month. Here, the supplies, diversions, and retardations are hardcoded to be the same as those for regulatory runs (in that the first month’s 50% supply, etc. is used each step of the forecast). Though results of a traditional six-month forecast run cannot be duplicated this way, month-to-month simulation of regulation decisions are faithful to the actual process. “1” indicates that Plan 1977-A’s internal supplies, retardations, and diversions should be used. This option reflects one variation evaluated in a sensitivity analysis for modifications to Lake Superior regulation practices. Example:
P77A COMPAT MODE: 0

SUP CRITERIA ROUND **2-3**
Indicates whether to round calculations regarding Criteria (b) & (c) (and determination of nominal non-gated flow) according to Plan 1977-A practices, or according to general rounding settings. “1” forces the criteria computations to be rounded according to P77ALEVELROUND, P77AFLOWROUND, and P77ABOMROUND (by default, four decimals for intermediate levels, whole numbers for intermediate flows, and beginning-of-month levels to the nearest cm). “2” allows the criteria calculations to be rounded to values established by the general rounding parameters . Example:
SUP CRITERIA ROUND: 2

SUP S.W. PIER FORMULA **2-4**
Indicates whether to use the old relationship for converting Lake Superior levels to S.W. Pier levels (“1”), or the new Environment Canada (2000) equation (“2”). Example:
SUP S.W. PIER FORMULA: 1

P77A MIDLAKES INC **2-5**
Positive integer number of sub-increments within each monthly routing interval for the middle lakes routing built into Plan 1977-A (optional). Similar function as "MIDLAKES INC", but applies only to middle lake routing within the Plan1977-A forecast computations. “40” is the traditional (and default) value. Example:
P77A MIDLAKES INC: 22

P77A STM RIV EQN, P77A STCRIV EQN, P77A DET RIV EQN, P77A NIA RIV EQN **2-6 to 9**
Discharge equation parameters to use with Plan1977-A (optional). The St. Mary's equation is used for the preproject relationship (Criterion (c) checking). The other three are used for Plan1977-A forecast routing and Criterion (b) checking. Same syntax as for other equation parameters (i.e., $Q=K(Z_u*Wt+(1-Wt)Z_d-YM)^A(Z_u-Z_d+Z_c)^B$). If not specified, the usual equations are used.
Examples:
P77A STM RIV EQN: 824.7, 181.43, 1.5, 0, 1,0
Here:

K	= 824.7	(outflow equation coefficient),
YM	= 181.43	(outlet bottom elevation),
A	= 1.5	(exponent for “depth” term),

B = 0 (exponent for “fall” term),
Wt = 1.0 (weighting factor for the upstream lake), and also
Q (discharge),
Zu (upstream water level),
Zd (downstream water level), and
Zc (fall adjustment constant).
P77A DET RIV EQN: 70.7, 165.94, 2.00, 0.50, 1.00, -.01
P77A NIA RIV EQN: 616.7 , 169.75, 1.50, 0.00, 1.00, 0 , 1953 Niagara Eqn

P77A FLOW ROUND, P77A LEVELROUND, P77A MONTH ROUND 2-10 to 12
Rounding parameters for Plan 1977-A (optional). Similar to general rounding parameters. Determines rounding for all level and flow calculations within Plan 1977-A and criteria checks (except the flow used in the U.S. Slip equation for Criterion (b) check, which is hardcoded to round to 10m3s). If not specified, rounding is to the nearest m3s, and the nearest 1/10 of a cm, except at the beginning of a month (cm). Examples:
P77A BOM ROUNDING: -2
P77A LEVEL ROUNDING: -4
P77A FLOW ROUNDING: 0

PREV STM RIV FLOW 2-13
The previous month's Lake Superior outflow (required for Plan 1977-A). The units must be one of the recognized flow units (i.e., 'm3s', 'cfs', etc). Example:
PREV STM RIV FLOW: 246 10m3s

SUP BOM LEVEL, MHU BOM LEVEL, MIC BOM LEVEL, HUR BOM LEVEL, STC BOM LEVEL, ERI BOM LEVEL 2-14 to 19
Beginning-of-month levels for Lakes Superior, Michigan-Huron, Michigan, Huron, St. Clair, and Erie (required for Plan 1977-A, if "START DATE" is not the first of a month; otherwise only “… START LEVEL” parameters are needed (see Section 4.3.5)). Examples:
SUP BOM LEVEL: 183.05 m
MHU BOM LEVEL: 579.3329 feet
STC BOM LEVEL: 174.3 m
ERI BOM LEVEL: 173.7 meters

SUP DEVIATION DATA 2-20
The amount of any deviations from the Lake Superior regulation plan. Appears as a "special" column in Lake Superior’s summary input file. If specified, legitimate values must be present for all time periods. Use 0 if no deviation for a given period. The units must be one of the recognized flow units (i.e., 'm3s', 'cfs', etc). Example:
SUP DEVIATION DATA: m3s, monthly, 12, 0, 0, 0, -300, 0, 0, 150, 150, 0, 0, 0, 0

SUP GATES DATA 2-21
The number of gates used for forced gate settings. Appears as a "special" column in Lake Superior’s summary input file. A value must be supplied for each time period (either a legitimate gate setting, or the missing data marker (-9999)). Legitimate values override the result of any regulation plan, but regulation plan results are calculated for reference. Valid settings are 0.5 and integers from 1 to 16. Units may be either 'NA' or 'None' or omitted. No

more than one decimal position will be used. Examples:
SUP GATES DATA: na, monthly, 4, -9999,-9999, -9999, -9999
SUP GATES DATA: 8, -9999, -9999,-9999, 0.5, -9999, -9999, -9999, -9999

SUP OUTFLOW DATA **2-22**
The total St. Mary's River flow used for forced flow settings. Appears as a "special" column in Lake Superior's summary input file. A value must be supplied for each time period (either a legitimate flow, or the missing data marker (-9999)). Legitimate values override the result of any regulation plan, but regulation plan results are calculated for reference. Units must be one of the recognized flow units (i.e., 'm3s', 'cfs', etc). No more than one decimal position will be used. **BUG**: Converts to m3s before looking to see if the value is –9999 (cfs), and fails to recognize it as missing, so (until fixed) please use m3s. Examples:
SUP OUTFLOW DATA: m3s, monthly, 4, -9999, -9999, -9999, -9999
SUP OUTFLOW DATA: 8, -9999, -9999, -9999, 1300, -9999, -9999, -9999, -9999

SUP SIDECHAN DATA **2-23**
The total non-gated flow for the St. Mary's River, which should be specified if forced gate settings are used (otherwise, the maximum sidechannel flow (2320 m3s) is assumed). Appears as a "special" column in Lake Superior's summary input file. A value must be supplied for each time period (either a legitimate flow, or the missing data marker (-9999)). Legitimate values override the result of any regulation plan, but regulation plan results are calculated for reference. Units must be one of the recognized flow units (i.e., 'm3s', 'cfs', etc). No more than one decimal position will be used. **BUG**: Converts to m3s before looking to see if the value is –9999 (cfs), and fails to recognize it as missing, so (until fixed) please use m3s. Examples:
SUP SIDECHAN DATA: m3s, monthly, 4, -9999, -9999, -9999, -9999
SUP SIDECHAN DATA: 8, -9999, -9999, -9999, 1300, -9999, -9999, -9999, -9999

4.3.3 Middle Lakes Routing Parameters

MIDLAKE SOLUTION **3-1**
Determines whether to use the iterative or matrix solution to solve nine simultaneous equations for the middle lakes. The iterative method is traditional, and keeps iterating until convergence or a user-specified tolerance is reached between successive answers. The matrix solution is the heart of GLERL's "MIDLAKES" program. If the first letter after the ':' is an 'M', then matrix algebra is used; if 'I', the iterative solution is employed. Examples:
MIDLAKE SOLUTION: iterative, (not to be confused with GLERL's matrix solution!)
MIDLAKE SOLUTION: matrix

ROUTING TITLE **3-2**
Description applicable to middle lakes routing (optional). Headings for input descriptions and output summaries. Maximum 79 characters. Example:
ROUTING TITLE: Routing through M-H, St. Clair, and Erie using GLERL's 1998 "MidLakes" model.

STM RIV EQN, STC RIV EQN, DET RIV EQN, NIA RIV EQN **3-3 to 6**
Discharge equation parameters for the upper connecting channels. The St. Mary's River equation is required for routing through Lake Superior using an outflow equation (e.g., the

preproject relationship). The St. Clair, Detroit, and Niagara River equations are required for middle lakes routing. Specify a comma-delimited list of 5-6 floating-point values: equation coefficient (K), channel invert (YM), depth exponent (A), fall exponent (B), upstream lake weighting factor (Wt), and fall adjustment constant (Z_c), where
$Q=K(Z_u*Wt+(1-Wt)Z_d-YM)^A(Z_u-Z_d+Z_c)^B$ is the general form of the equation. If a comma terminates the six required fields, then the rest is available for comments. Examples:
STC RIV EQN: 84.1168, 543.4, 2.00, 0.50, 0.50, 0, English 55
DET RIV EQN: 128.0849, 543.81, 2.00, 0.50, 1.0, 0
NIA RIV EQN: 260.5, 550.11, 2.20, 0.00, 1.0,0 ,English 55
STC RIV EQN: 46.4, 165.83, 2.00,0.50, 0.50, -0.02
DET RIV EQN: 70.7, 165.94, 2.00, 0.50, 1.00, -.01
NIA RIV EQN: 182.78 , 167.24, 1.66666666666667, .5, .6, 0, Metric Boyd eqn

CGIP AVG LEVEL DATA, CGIP STARTLEVEL **3-7 to 8**
Stage to use for the Niagara River's Chippewa-Grass Island Pool, which serves as the downstream level in two-gage discharge relationships for the river. "CGIP AVG LEVEL DATA" is required if a two-gage Niagara equation is used. An IJC directive sets this level at171.16m IGLD-85. Appears as a "special" column in Lake Erie's summary input file. "CGIP STARTLEVEL" is needed if a two-gage Niagara equation is used with the matrix method (though future versions may assume a value equal to the period average level).
Examples:
CGIP AVG LEVEL DATA: Monthly, Constant, 171.16 m
CGIP START LEVEL: 171.16 m

4.3.4 Lake Ontario Regulation Parameters

Since the Lake Ontario/St. Lawrence River module is currently undergoing testing, this is NOT presently a comprehensive listing.

U STL RIV EQN **4-1**
Describes how to calculate Lake Ontario outflows (required if routing through Lake Ontario). If '1958D', then it uses Plan 1958D. Example:
U STL RIV EQN: 1958D

DPR RIV FLOW DATA, MII RIV FLOW DATA **4-2 to 3**
The amount of Des Prairies and Milles Isles Rivers flow. Appear as "special" columns in Lake Ontario's summary input file. If specified, legitimate values must be present for all time periods. The units must currently be 'm3s'. Example:
DPR RIV FLOW DATA: ...\data\dpmiqflo.df

ONT DEVIATION DATA **4-4**
The amount of any deviations from Lake Ontario regulation Plan 1958-D. Appears as a "special" column in Lake Ontario's summary input file. If specified, legitimate values must be present for all time periods. Use 0 if no deviation for a given period. The units must currently be 'm3s'. Example:
ONT DEVIATION DATA: m3s, qmonthly, 16,0,0,0,-300,-250,-250,-150,-150,0,0,0,0,0,0,0,0

ONT DEVIATION DATA: 16, 0, 0, 0, 0, 0, 0, 50, 100, 90, 0, 0, 0, 0, 0, 0, 0
ONT DEVIATION DATA: 16, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0

LS ROUGHNESS DATA   4-5
Coefficients for Long Sault roughness data, which appears as a "special" column in Lake Ontario's summary input file. If specified, legitimate values must be present for all time periods. Units may be either 'NA' or 'None'. No more than six decimal positions will be used.
Example:
LS ROUGHNESS DATA: ...\data\lsnqm.df

4.3.5 Miscellaneous Parameters

SUP AREA-ELEV, MHUAREA-ELEV, MIC AREA-ELEV, HUR AREA-ELEV, STC AREA-ELEV, ERI AREA-ELEV, ONT AREA-ELEV   5-1 to 7
Area vs. elevation data for Lakes Superior, Michigan-Huron, Michigan, Huron, St. Clair, Erie, and Ontario. Routing through the middle lakes requires values for "STC AREA-ELEV", "ERI AREA-ELEV", "MHU AREA-ELEV" ("MIC AREA-ELEV" and "HUR AREA-ELEV" are reserved for future use, when precipitation or evaporation might be specified in terms of inches or cm). Routing though Lake Superior requires "SUP AREA-ELEV" (and if using Plan 1977-A, data for the middle lakes is required as well). Routing though Lake Ontario requires only "ONT AREA-ELEV". If only one record is specified for a lake, the area is assumed constant, and no elevation is required. If more than one record is specified, each record is taken as a point on the elevation-area curve (maximum 5 points). To replicate existing practice (using uniform month length), the areas below backed out of "storage constants" must be used.   Values backed out of "storage constants" for converting monthly flows to lake heights (month length=2629746 s):

Lake	"Storage Number"	Area (square meters)
Sup	31380	82521429480
MH	44670	117470753820
StC	427	1122901542
Erie	9770	25692618420

If the end of the string is comma delimited, the rest of the line is available for comments.
Examples:
SUP AREA-ELEV: 82521.429480 km2,   (from "storage constant")
MHU AREA-ELEV: 117470753820 m2
MIC AREA-ELEV: 6.2169D11 m2
HUR AREA/ELEV : 6.4120D5 km2
STC AREA ELEV: 1198.8 km2
ERI AREA-ELEV.: 9913 mi2, 579 feet,   (these points estimated
ERI AREA,ELEV : 9900 mi2, 569 feet,   from depth-volume curve in
ERI AREA-ELEV : 9889 mi2, 559 feet,   Coord. Phys. Data book)

SUP START LEVEL, MHU STARTLEVEL, STC START LEVEL, ERI START LEVEL, ONT START LEVEL   5-8 to 12
Initial lake levels for Lakes Superior, Michigan-Huron, St. Clair, Erie, and Ontario (i.e., for date given in "START DATE"; required if routing through the lake). Elevations must be expressed in IGLD-85. If the end of the units string is comma delimited, then the remainder of the line is

available for comments. Examples:
SUP START LEVEL: 183.05 m,   (estimated)
MHU START LEVEL : 579.3329 feet
STC START LEVEL: 174.3 m

SUP TIMESTEP, MIDLAKE TIMESTEP, ONT TIMESTEP 5-13 to 15
Indicates routing interval to use for Lakes Superior, Michigan-Erie, and Ontario (required if routing through the lake). Must be "D", "W", "Q", or "M" (only the first letter is necessary).
Examples:
SUP TIMESTEP: monthly
MIDLAKE TIMESTEP   :   q
ONT TIMESTEP: week (only sees the 'w')

MIDLAKE INC, P77A MIDLAKE INC, SUP INC, ONT INC 5-16 to 19
Positive integer denoting the number of sub-increments within each routing interval. If the first character after the ':' is an 'H', 1-hour increments are used (in the case of hourly increments, actual month length, and monthly or quarter-monthly routing, the number of increments varies each month or quarter-month, and is adjusted automatically). This parameter has a significant effect on routing mathematics. If the modeler wishes to assume outflow constant throughout each routing interval (traditionally assumed on Lake Superior when using Plan 1977-A), then one increment must be specified. If the outflow will be modeled as a function of level (as in the middle lakes, or on Lake Superior if the routing interval is treated separately from the regulation period), then the storage change during a given routing interval is integrated over the specified number of increments. If too small, the solution can fail, especially if there are large, volatile retardation flows involved. Besides influencing the accuracy of integration over an entire routing interval (if using the matrix method) it also impacts the accuracy of solving the simultaneous equations for any given increment. It is recommended to use increments of 222,000 s or less (about 2.5 d). Thus, use a minimum of 12 increments for monthly routing and at least four for weekly or quarter-monthly. A single increment can be used for daily routing. These suggestions are to be considered minimums since anything less tends to result in inaccuracies noticeable at the centimeter level. Traditional practice employs 10 increments in a quarter-monthly routing and 40 in a monthly routing. Although there is no mathematical downside to using many increments, at some point, the returns diminish to the point of making no discernible difference (e.g., it is estimated that the CGLRRM's 64-bit floating point (double precision) calculations do not benefit from more than 100 increments per month). Additionally, depending on log file settings, using hundreds of increments can seriously affect performance. Note that setting the number of increments such that they are each one hour long far exceeds mathematical requirements but it is a safe setting for any purpose. Example:
sup Inc : 1
MIDLAKE INC: hourly

SUP NBS DATA, MHU NBS DATA, MIC NBS DATA, HUR NBS DATA, STC NBS DATA, ERI NBS DATA, ONT NBS DATA 5-20 to 26
Repeating data or filenames for net basin supply data for Lakes Superior, Michigan-Huron, Michigan, Huron, St. Clair, Erie, and Ontario. If not provided, warning messages are triggered, but computations proceed assuming zeros. Only the first word after the ":" is used; the rest is available for comments. Specify a valid file specification or repeating data parameter line(s).

Examples:
SUP NBS DATA: ...\indata\spmnbs.boc (Lakes Superior NBS)
ERI NBS DATA: NULL
HUR NBS DATA: /wcds/coordata/nbs/humnbs.cor

SUP CON USE DATA, MHU CON USEDATA, MIC CON USE DATA, HUR CON USE DATA, STC CON USE DATA, ERI CON USE DATA, ONT CON USE DATA 5-27 to 33

Repeating data or filenames for consumptive use and "other supplies" for Lakes Superior, Michigan-Huron, Michigan, Huron, St. Clair, Erie, and Ontario. As with the precipitation, runoff, and evaporation components, historical consumptive use is usually already built into the most common NBS data set, computed using the residual method. If a consumptive use file is specified for a given lake in addition to the NBS file, ensure that the data reflect not absolute magnitudes, but relative adjustments to NBS values. Specify a valid file specification or repeating data parameter line(s). Examples:
HUR CON USE DATA: …\indata\spmcon.boc
MIC CON USE DATA: /wcds/histdata/spmcon.cor
MHU CON USE DATA: NULL

SUP OTHER SUP DATA, MHU OTHER SUP DATA, MIC OTHER SUP DATA, HUR OTHER SUP DATA, STC OTHER SUP DATA, ERI OTHER SUP DATA, ONT OTHER SUP DATA 5-34 to 40

Repeating data or filenames for other supplies for Lakes Superior, Michigan-Huron, Michigan, Huron, St. Clair, Erie, and Ontario; to provide flexible input. Specify a valid file specification or repeating data parameter line(s). Examples:
HUR OTHER SUP DATA: …\indata\spmgdw.boc
MIC OTHER SUP DATA : /wcds/histdata/grndwat/spmgdw.cor
MHU OTHER SUP DATA: NULL

LONG LAC DIV DATA, OGOKI DIV DATA, COMB LLO DIV DATA 5-41 to 43

Repeating data or filenames for the Long Lac and Ogoki diversions into Lake Superior. At least one of these data is needed if routing through Lake Superior. Absence triggers warning messages, but computations will proceed assuming zeros. Specify a valid file specification or repeating data parameter line(s). Examples:
LONG LAC DIV DATA: …\indata\spmllk.tmp
COMB LLO DIV DATA: m3s, monthly, 12, 113,99,85,79,181,252,207,170,133,127,122,125
OGOKI DIV. DATA: NULL
COMB LLO DIV DATA: /wcds/coordata/diversions/spqllo.cor

CHICAGO DIV DATA, WELLAND CANAL DATA 5-44 to 45

Repeating data or filenames for the Chicago diversion out of Lake Michigan and the Welland Canal discharge from Lake Erie. These parameters are needed for middle lakes routing and Welland data is expected if routing through Lake Ontario. Absence triggers warning messages, but computations proceed assuming zeros. Neither is used by Plan 1977-A, which has its own internal values. Specify a valid file specification or repeating data parameter line(s). Examples:
CHICAGO DIV DATA: 10m3s, constant, -9.1
CHICAGO DIV. DATA : …\indata\mhmchi.act
CHICAGO DIV DATA: NULL

WELLAND CANAL DATA: /wcds/coordata/runoff/erqwel.cor
WELLAND CANAL DATA: m3s, monthly, 3,-218,-215,-218
WELLAND CANAL DATA: 9, -252,-255,-255,-249,-252,-258,-261,-258,-227

SUP OTHER DIV DATA, MHU OTHER DIV DATA, MIC OTHER DIV DATA, HUR OTHER DIV DATA, STC OTHER DIV DATA, ERI OTHER DIV DATA, ONT OTHER DIV DATA **5-46 to 52**
Optional repeating data or filenames for other diversions for Lakes Superior, Michigan-Huron, Michigan, Huron, St. Clair, Erie, and Ontario; to provide flexible input. Specify a valid file specification or repeating data parameter line(s). Examples:
ERI OTHER DIV DATA: ...\indata\erqdiv.brl
MIC. OTHER DIV. DATA: /wcds/crandonmine/grndwat/mhddiv.99
MHU OTHER DIV DATA: NULL

STM RIV FLOW DATA, STC RIVFLOW DATA, DET RIV FLOW DATA, NIA RIV FLOW DATA **5-53 to 56**
Optional repeating data or filenames for the flow in the upper connecting channels. Added to the total water balance for each lake, as if another NBS or diversion term. Normally," STC RIV FLOW DATA" or "DET RIV FLOW DATA" are not required, unless simulating regulation of the middle lakes. Use "STM RIV FLOW DATA" to explicitly specify outflow from Lake Superior, instead of using the preproject equation or a regulation plan (not implemented yet). Use "STM RIV FLOW DATA" or "NIA RIV FLOW DATA" to explicitly specify flows in these rivers, if routing through and solving middle lakes but not Lake Superior, or routing through Lake Ontario but not the upper lakes. Specify a valid file specification or repeating data parameter line(s). Examples:
NIA RIV FLOW DATA: quarter-monthly, tcfs, 8, 230, 240.5, 240, 255, 234.4, 255, 234, 222
STM RIV. FLOW DATA: ...\tsdata\smdflo.try
NIA RIV FLOW DATA: 10, 213, 214.5, 210, 255,224, 215.5, 234, 220, 215, 217

STM RIV ICE DATA, STM RIV WEEDDATA, STM RIV RETR DATA, STC RIV ICE DATA, STC RIV WEED DATA, STC RIV RETR DATA, DET RIV ICE DATA, DET RIV WEED DATA, DET RIV RETR DATA, NIA RIV ICE DATA, NIA RIV WEED DATA, NIA RIV RETR DATA, STL RIV ICE DATA, STL RIV WEED DATA, STL RIV RETR DATA **5-57 to 71**
Optional repeating data or filenames for ice retardation, weed retardation, and combined ice/weed retardation for the St. Mary's, St. Clair, Detroit, Niagara, and St. Lawrence Rivers. Specify a valid file specification or repeating data parameter line(s). Examples:
DET RIV RETR DATA: ...\indata\scqciw.tmp
STC RIV WEED DATA: /wcds/data/test/mhdweed.023

SUP EOP LEVEL DATA, MHU EOP LEVEL DATA, STC EOP LEVEL DATA, ERI EOP LEVEL DATA, ONT EOP LEVEL DATA **5-72 to 76**
Optional repeating data or filenames for forced end-of-period levels. Appear as "special" column in the summary input files. If specified, legitimate values must be supplied for each time period (a legitimate lake level or the missing data marker (-9999)). Legitimate values override the result of any regulation plan, but the regulation plan results are calculated for reference.   The units must be one of the recognized distance units (i.e., 'm', 'feet', etc). **BUG:**

Program converts to meters before looking to see if the value is –9999 (feet), and fails to recognize it as missing, so (until fixed) please use units of 'm'. No more than two decimal positions will be used. The computations for the current routing interval are not affected - as these parameters also specify forced BOP levels for the next period (Plan 1977-A requires BOP levels for the next period on the middle lakes). Specify a valid file specification or repeating data parameter line(s). Examples:
ERI EOP LEVEL DATA: quarter-monthly, m, 8, -9999, -9999, -9999, -9999, -9999, -9999, -9999, -9999
STC EOP LEVEL DATA: 12, -9999, 174.30, -9999, -9999, -9999, -9999, -9999, -9999, -9999, -9999, -9999, -9999

SUP SUM INPUT FILE, MHU SUM INPUT FILE, STC SUM INPUT FILE, ERI SUM INPUT FILE, ONT SUM INPUT FILE 5-77 to 81

Optional summary files from which to read time-series input data. These, follow a specific format, and are built from the time-series data inputs, using names like "weekly.stc", "monthly.sup", or "daily.m-h", based on the lake and routing intervals. They can be reused if the time-series data remains the same (e.g., Say, runs of a sensitivity analysis looking at the impact of different supplies on Lake Superior. Given monthly routing on Lake Superior and quarter-monthly routing on the middle lakes, after the first run, the model will generate summary input files named "monthly.sup", "qmonthly.m-h", "qmonthly.stc", and "qmonthly.eri". Only the contents of the file for Lake Superior would vary between runs). Reuse by specifying the file names in the parameter statement. Files may be renamed, as desired. The various time-series data parameters (NBS, etc) for the middle lakes can be commented out. Appreciable time may be saved, especially if some of the input data is read from files (e.g., 90 years of historical supplies for each lake). Examples:
MHU SUM INPUT FILE: ./mhspecial.save
MHU SUM. INPUT FILE: C:\CGLRRM\TSDATA\MHINSUM.PL1
STC SUM INPUT FILE : qmonthly.stc

4.4 Sample Input File

The following illustrative sample input file is for a relatively short simulation through the upper lakes (owing to some humorous, fictional settings, results are fiction).

All comment lines must begin with a number sign (“#”). All other lines contain one of the parameter ID codes listed in Section 4.3, a colon (delineating it from the input parameters/data), the input parameters or data as prescribed in Section 4.3, and possibly a comment area.

Note the high degree of internal documentation. It is highly recommended that all users attempt to retain this level of documentation for all modelling efforts.

```
# SAMPLE FILE (modified version of C:\DEMO\DEMO.DAT)
# This is a large example parameter file to show recommended input architecture.
# The lines can be in any order.
#
# Set a moderate level of logging while reading these parameters
General Verbosity:      5
#
```

```
######################################################################
#######################     Run Control Parameters  ######################
######################################################################
#
# Route through upper lakes only
Lakes to Solve For:        SMCE
#
# Perform a simulation for approximately 29 months
Start Date      :          1990, 3, 23
End Date        :          1992, 8, 17
#
# Commented out the destination output file location; they will end in the PWD.
#output directory:         out\
#
# Name output files with a ".dem" at the end
Output Extension:          .dem
#
# Identify who created these results
Modeler Name:              G. Lake, III
#
Title 1:                   Demonstration Run
Title 2:                   May as well write something here
Title 3:                   May as well write something here, too!
#
#####################################################################
###################   General Routing Parameters     ####################
#####################################################################
# Rounding values (expressed in exponents).  -1 rounds to the  tenth; 2 rounds to the nearest
# hundred; 0 rounds to integer.
# If these lines are commented out/absent, default is not to round any intermediate values.
#
# These two lines control how answers are rounded within intermediate stages of the routing
# computations on all the lakes.  They provide compatibility with traditional practices.
Level Round :              -99  (do not round levels in intermediate computations)
Flow Round  :              0     (round intermediate flows to nearest 10 m3s)
#
# Existing practice rounds to centimeter when starting a new month, whether monthly or
# quarter-monthly routing.
BOM Round   :              -2  (round beginning-of-month levels to cm)
#
Routing Title:             Another title line, intended for middle lakes use, so write something!
#
# Existing practice uses concept of "uniform" month length, where February is same number of
# seconds as January.  This is default.  But lets use real time for the run.  Note that historical
# NBS's developed using uniform month length may be different than NBS recomputed using
# actual month length, so may want to adjust input values if using "actual".
Month Length: Actual
#
####################################################################
#################   Lake Superior Routing Parameters     ###############
####################################################################
#
# Specify a typical level to start with
Sup Start Level        :   183.4 m
#
# Specify routing interval (only first letter matters)
Sup Timestep:              qmonthly
```

```
#
# How many increments per interval.  Can be 1 if assuming constant outflow for period (linear
# change-in-storage), such as we do with Plan 1977-A.
Sup. Inc: 15
#
# Use constant coordinated Lake Superior surface area
Sup Area/Elev :           82100 km2
#
# Specify how to compute outflows.  Instead of Plan 1977-A, use outflow equation (1887
# relationship).
St. M. Riv. Solution:     outflow equation
#
St. M. Riv. Eqn:          824.7, 181.43, 1.5, 0, 1,0
#
# Which results to write to a time-series file.  For weekly routing, can have 'w', 'q', or 'm', or any
# combination.  Here, request weekly and quarter-monthly
Sup Output Files:         mq
#
# This file is missing values except for 183.36 m on 7 Dec. 1991.  This causes the BOP level for
# 8 Dec. 1991 to change to 183.36 m.
Sup EOP Level Data:       ...\data\forcelev.sup
#
# This file specifies 2000 m3s for 8 Dec. 1991
Sup Deviation Data:       ...\data\spddev.txt
#
###################################################################
##################   Middle Lakes Routing Parameters    ################
###################################################################
# Do daily routing on middle lakes
Midlake Timestep:         daily
MIDLAKE inc:              h  (i.e., 24 increments per interval)
#
M-Hu Start Level:         177.5 m
St. C Start Level:        175.80 m, IGLD 85, of course
Eri Start Level:          174.80 m
#
# Use the standard St. Clair River and Detroit River equations
# Q=K*(Zu*Wt+(1-Wt)Zd-YM)**A*(Zu-Zd+Zc)**B
#                           K       YM      A     B     Wt    Zc
St. C. Riv Eqn:           46.4,   165.83, 2.00, 0.50, 0.50, -0.02
Det. Riv. Eqn:            70.7,   165.94, 2.00, 0.50, 1.00, -0.01
#
# Boyd's two-gage Niagara Eqn: (minor correction by Quinn/Noorbakhsh not included)
Nia. Riv. Eqn:            182.78, 167.24, 1.66666666666667, 0.50, 0.60, 0
#
# GLERL's MIDLAKES model uses the matrix algebra method to solve the simultaneous
# equations on the middle lakes.  Traditional practice uses an iterative procedure.  Either way,
# should arrive at the same solution.  If using matrix, specify an initial CGIP level if using two-
# gage Niagara eqn.
Midlake Solution:         Matrix
CGIP Start Level:         171.16 m, (initial Chippewa/Grass Island Pool level)
#
# Request weekly, quarter-monthly and monthly output files for L. Erie
Eri Output Files:         qmw
#
# Request daily and monthly output for L. Michigan-Huron
Mhu Output Files:         dm
```

```
#
# Get no output for L. St. Clair (since commented out!)
#St. C. Output Files:      m
#
# One area parameter means constant area assumed
M-Hu  Area/Elev :          1.1747075382D11 m2
St. C.  Area/Elev :        1122901542D0 m2
#
# Multiple area/elevation pairs define a curve to interpolate from
# These are crudely estimated from book of coordinated data
Eri   Area/Elev :      9900 mi2, 569 feet
Eri   Area/Elev :      9913 mi2, 579 feet
Eri   Area/Elev :      9889 mi2, 559 feet
#
###################################################################
######################     Input Data Files        ######################
###################################################################
# Specify files to use for the various data types (monthly data).  Note the type of slash used
# between directories (Windows='\'; Unix='/').
#
# Repeating L. Superior NBS data (historical means 1900-1989)
Sup NBS data:              tcfs, monthly, 8, -14, 10, 45, 149, 186, 158, 130, 101
# Specify remaining four months on another line to keep first line short (note differences)
Sup NBS data:              4, 73, 38, 18, -24
#
# Read Long Lac/Ogoki diversion data from an external file
Comb LLO Div Data:     …\data\spmllom.cor
#
# Repeating L. Michigan-Huron NBS data (historical means 1900-1989) on one line
M-Hu NBS data: m3s, monthly,12,1494,2478,5220,8091,7096,5783,3604,1559,875,25,1030,809
#
# Chicago Diversion data
Chicago Div. Data:        10m3s, constant, 9.1
#
# Repeating L. St. Clair NBS data (historical means 1900-1989)
Stc NBS data: cfs,monthly,12,5000,7000,9000,8000,5000,3000,3000,1000,1000,1000,2000,5000
#
# L. Erie NBS data in a file:
Eri NBS data:               …\data\erqnbs.jb
# CGIP level assumed constant
CGIP Avg Level Data:   m, c, 171.16
#
# Pretend that there are strange things happening on the Welland (negative means flowing into
# Lake Erie).  Note the 48 values to be repeated each year may be spread across several lines.
Welland Canal data: m3s, qmonthly, 5,222, 111, 876, 252, 255,
Welland Canal data:   7,  275, 249,  252, 258,  261,  258,  543
#  Had a little gravity problem late in April:
Welland Canal data: 12, 218, 321, -17, 132, 119, 255, 386, 252, 258, 261, 157, 177
Welland Canal data: 12, 218, 215, 147, 252, 255, 255, 74, 184, 98, 261, 258, 174
Welland Canal data: 12, 218, 217, 265.987, 73, 579, 94, 115, 392, 258, 261, 471, 101
#
# Miscellaneous made-up data
#
# Fleets of freighters head for Asia …
Sup Other Div data:       m3s, mon, 12, 0,0,0, -19, -15, -8, -7, -8, -10,-12, -14, 0
#
# … but first, they empty their ballast before heading down the St. Clair River
```

```
M-Hu Other Div data:     m3s, mon, 12, 0,0,0, 19, 15, 8, 7, 8, 10,12, 14, 0
#
# Beer brewing hits the Midwest by storm
Sup Con Use data:        tcfs, constant, 1
#
# Steel mills make a comeback in Indiana
Mic Con Use data:        cfs, constant, 6.5
#
#  A mine pumps groundwater into the Wolf River
Mic Other Div data:      ...\data\ontdev.prn
#
#  Hydrogeologists locate a groundwater sink off Tobermory
Hur Other Div data:      10m3s, constant, -45.2
#
# New farms pop up all over the eastern shore of Lake St. Clair
St. C. Con Use data:     tcfs, constant, 1
#
#  Ice/Weed retardation for St. Clair, Detroit, and Niagara R.
St. C Riv Retr. data:    m3s, monthly, 12, 590, 480, 110, 110, 0, 0, 0, 0, 0, 0, 0, 110
Det Riv retr data:       m3s, monthly, 12, 650, 510, 110, 0, 0, 0, 0, 0, 0, 0, 0, 140
Nia Riv retr data:       m3s, monthly, 12, 110, 140, 80, 140, 0, 60, 230, 140, 80, 60, 0, 0
#
################################################################
#################       Run-Time Message Level      ##################
################################################################
#
# The last verbosity level read remains in effect during execution, and determines the amount of
# output written to the log files.
TS Data Verbosity:       8
Sup Verbosity:           5
MidLake Verbosity:       3
Check Verbosity:         7
General Verbosity:       3
```

5 PROCESSING

The computational core of the CGLRRM is straightforward, highly portable, and easy to build, providing consistent results on a variety of platforms. It is a single-threaded application that spawns no new processes. It is statically linked (i.e., no run-time libraries or DLL's required), and does not interact with any external processes (or even the console). The only system functions used are the most basic file input/output (I/O) routines and a non-essential call for the system time/date. Essentially, it simply reads and writes files to disk, processing data using 64-bit floating-point math (and only real numbers). Processing moves sequentially through four separate phases: initialization, times-series data pre-processing, routing computations, and post-processing.

A primary source of undefined situations and confusion has been the option of allowing four different time intervals to co-exist. Time-series input data maybe given as monthly, quarter-monthly, weekly, daily, or constant. The time interval of the input data is independent of the timestep used for routing computations; the values are translated to the necessary routing interval automatically using simple procedures. Monthly input data used in a quarter-monthly routing result in the same number being repeated four times. Daily input data gets averaged over blocks of seven or eight values, in order to be used in a quarter-monthly routing.

Conversions involving weekly data or routing intervals use a straightforward weighted-average. The routing interval may be monthly, quarter-monthly, weekly, or daily. This is the basis of the change-in-storage computations, defining the BOP, EOP, and timestep. The basic results of the routing computations are levels and flows, with individual values for each routing interval. The model also computes output for periods of greater length than the routing period (e.g., given a weekly routing, quarter-monthly and monthly averages of the weekly levels and flows are derived). Whether or not these results are written to an output file is user-defined.

Finally, the third type of time interval is the regulation period. The CGLRRM accommodates time periods for regulation and routing, such as doing monthly regulation (i.e., Plan 1977-A) on Lake Superior, yet routing on a weekly timestep (the regulated flow is computed whenever a week starts a new month, and assumed constant for the month, resulting in weekly levels and flows). This concept complicates matters significantly, and many issues must still be addressed.

5.1 Initialization

First, the detailed log file (noting information such as run start time and program version/date) is opened. This goes into the present working directory when the program starts, and logging begins immediately. Next the master input file is opened and input parameter files are read in the order they are listed. File I/O actions and results are logged, as well as feedback regarding the parameters being processed. After reading all the parameters supplied, the program performs a "consistency-check". This module tests the input parameters for completeness, extra data or parameters, and mismatches, and logs the results (e.g., a message is generated if a run fails to supply values for Lake Michigan diversion at Chicago). It will flag the user if only the upper lakes are being solved but Lake Ontario supplies are given (extra data). It also logs messages describing any automatic conversions (e.g., converting a diversion flow given in cfs to tens of m3s). Users are urged to pass constructive comments regarding this message database along to the model custodian.

5.2 Time-Series: Pre-Processing of Data

By limiting the number of routing periods available, the various input data files (NBS, diversions, etc.) could be opened one-by-one and entire contents read in. This is the most common approach, with the fastest execution. But it also imposes a minimum memory requirement, and fixes limits that tend to spawn alternate versions on specific systems, for specific uses. Another approach is to leave the files open during processing, and just read the next line as needed. However, the number of files that would need to be open simultaneously for a run through the total system, would sometimes exceed the default number of files allowed open on older systems (typically 20-40). Therefore, a third alternative was adopted: converting and condensing all the time-series routing data into summary files.

One-by-one, all the time-series data files for each lake are summarized through the entire simulation period (e.g., for a quarter-monthly simulation of the middle lakes, using default filenames, the CGLRRM writes three intermediate files in the present working directory: qmonthly.stc, qmonthly.m-h, and qmonthly.eri). These summary inflow files follow a strict format. They can be saved under a new name, and reused in other runs by specifying the new name as a parameter. This saves effort if multiple runs are made wherein the water balance and flow retardation terms don't change (e.g., such as for optimizing a regulation parameter).

Writing these intermediate files requires disk activity that wouldn't be necessary if all the data could be held in memory, or the source data files read in, as the routing computations required. However, it avoids hard limits on memory requirements and the number of simulation periods per run, without requiring reconfigurations. For short simulations of a few hundred periods, performance degradation is minimal. Additionally, for certain long jobs (e.g., optimization studies), some or all of the summary inflow files are reusable. Another tradeoff is that values of individual components are not readily available during routing computations. However, the routing computational code is simplified, since all checking, converting, and interpolating regarding the various time-series data has already been done.

5.3 Routing Computations

The routing computations are performed in a loop directed by file, "CONTROL.FOR". For each simulation period, the routine takes the beginning-of-period water levels, etc., and the water balance/retardation terms, and runs them through the specified regulation plans and routing mechanisms. Results for each simulation period get written to files in the output directory. The general structure consists of three modular "loops": Lake Superior, the middle lakes, and Lake Ontario/St. Lawrence. Lake Superior outflows can be determined using an outflow equation or Plan 1977-A (the 1955 Modified Rule of '49 will be added later). The middle lakes can be solved via iterative or matrix methods. Should Lake Erie regulation be desired in the future, it can be added as an alternative routing method. The Lake Ontario/St. Lawrence portion is being implemented, and links regarding the input, output, and control logic are incorporated.

The control loops are nested so that the most downstream lake is the outer-most loop. For instance, in system-wide runs, the outermost loop is Lake Ontario’s. The program checks to see if Niagara River flows are available throughout the current Ontario routing period. If so, it executes the Ontario module to route through the period, and repeats this process until the Ontario period ends later than the available Niagara flows. Then it jumps to the next inner loop, and attempts to execute the middle lakes module until enough Niagara River flows are computed to perform the next Ontario routing. If the St. Mary's River flows end before the current middle lakes period, it jumps to the innermost loop, and routes through Lake Superior until the date passes the end of the current middle lakes period.

6 OUTPUT

In general, primary output consists of resulting levels and flows, and pertinent regulation plan information. Statistics, summaries, and specialized reports must currently be derived manually, as desired. Such features may be implemented in future versions, and could be made part of the “back-end” graphical user interface.

6.1 Flexibility

In order to maximize the flexibility of output details, an innovative approach was used. A family of parameters dubbed “Verbosity” (literally, “an expressive style that uses excessive words”) was incorporated which permits user-defined control of the level of detail desired during each critical processing step. Full details are given in Section 4.3.1. These parameters effectively control the degree of debugging information written to either the screen or the detailed log file. Values may range from 1 to 9 (most lax to most comprehensive). If not provided, these parameters default to a general setting of 5 to provide limited, yet sufficient information for most practical purposes.

Users may also provide an (optional) extension for the program to attach to all output files for a specific run as well as the output directory to write all files to for that run, to help maintain files in an orderly fashion. Again, full details are provided in Section 4.3.1.

Finally, the user must define which optional output files are desired for each lake in addition to the standard files. Daily, weekly, quarter-monthly, and monthly files may be requested, or any combination of these as desired (note that the routing interval must be longer than the computation interval(s) requested, however). For example, it is possible to request output files for Lake Erie of monthly, quarter-monthly, and weekly values from a run using a weekly routing period, but not files of daily values.

6.2 Limitations

Logically, one might expect a post-analysis package to be readily available for direct use with such an extensive model development. However, since it was desirable to use Fortran 77 coding exclusively, design of a post-analysis package proved too complex to internalize with Fortran code. Currently, macros are used to loop the routing model. Any probabilistic and statistical analyses must be performed externally using separate software packages. Any specialized summaries and reports (such as those provided by many earlier models) must also be derived manually at present. Nonetheless, many features such as these will likely be implemented when the back-end GUI is created for use with future versions.

6.3 Description of Standard Output Files

As well as the large number of optional output files that may be generated after each run of the model (Section 6.4) there are a significant number of standard output files that are written. The following is a comprehensive listing of these files, along with a brief description of each.

New users are urged to browse each of the files below in the example directory “SUPREG” before proceeding with model run attempts.

DETAILED.LOG

Designed as an aid for debugging, this extensive file is a log created during CGLRRM processing. The level of detail is controlled using the “Verbosity” family of input parameters (see Section 4.3.1). It lists the run time and date, as well as the program version and compilation information. Next, it echoes comments and parameter settings from each of the input files (in the order they are listed in file CGLRRM.INI). Any relevant warning or error messages are logged at the point where any perceived problem was found. Then, each processing step is logged as it is performed, and again, any error or warning messages are written. Details of each computational step are included, such as data values, results, trial settings, etc. A summary of errors found during computations is given. Finally, a summary of all input parameters and comments is provided. It should be possible to reproduce the results of any particular run by using this final section as an input parameter file (provided the same external time-series data files are available in the same locations, and excepting differences due to model versions).

This file is written to the present working directory at run time.

FAILURE.LOG

This output file contains a comprehensive list of all warning and error messages logged during runs of the model. It is important to note that this file is appended to during each subsequent run of the program, so it is imperative to delete this file before each run if the user wishes to view a log of messages pertaining only to the last run performed. This file was designed to be appended to in lieu of being rewritten each run in order that a full list of messages could be logged during the entire debugging sequence.

This file is written to the present working directory at run time.

SUMMARY.???

This file is written to the output directory that the user defines, and will have the same extension as the user defines for optional output files (Section 4.1). It serves to mirror all input data in a comprehensive listing. Standard headings are included to help the user differentiate between the various types of input used for a particular run (and to permit this file to be used as an input file to recreate runs provided that associated data files are available and the model version is the same). In the future, it is intended that an executive summary of each run will be added at the top of this file, along with some simple statistics (i.e., mean level during the simulation, standard deviation of the flows, etc.).

SUPERIOR.???

This file is written to the output directory that the user defines, and will have the same extension as the user defines for optional output files (Section 4.1). This file is written only for Lake Superior regulation runs, and details the routing period and number of increments, as well as the six-month forecasted levels (beginning- and end-of-period, mean, and Southwest Pier), flows (balancing equation, plan, preproject, actual, gated, and sidechannel) and the gate setting and requirement codes.

MICHERIE.???

This file is written to the output directory that the user defines, and will have the same extension as the user defines for optional output files (Section 4.1). This file details the routing through the midlakes (if used). Included are the routing interval, number of increments, stage-discharge equation parameters, routing period, as well as a summary of mean and end-of-period levels for each lake and mean outflows through each of the connective channels (St. Mary's, St. Clair, and Niagara Rivers).

SP?TSD.???, MH?TSD.???, SC?TSD.???, and ER?TSD.???

This file is written to the output directory that the user defines, and will have the same extension as the user defines for optional output files (Section 4.1). The first "?" denotes the routing period chosen for each particular lake (see Section 4.1). Each of these files contains an inflow/outflow summary for time-series data for Lakes Superior, Michigan-Huron, St. Clair, and Erie, respectively. Though each of these files varies somewhat from the next, they are generally of similar format. Included are any user-defined time-series data, such as net inflows and retardations, end-of-period levels, ice/weed/retardation data, outflows, net basin supplies, diversion/consumptive use/other supply data, as well as number of gates, sidechannel flow, and deviations for Lake Superior, and the Chippewa-Grass Island Pool levels for Lake Erie.

6.4 Formats of Optional Output Files

Due to the vast number of optional output files that may be generated from any one run, it is not feasible to describe each possible format in great detail here. In general, most of the user-defined time-series output files follow the specific coordinated formats also used for time-series input files (see Section 4.1 for details).

Optional output files contain only ASCII characters within the range from 32-126 (base 10), with a carriage return and/or line feed terminating each record. A value of -9999 indicates missing data. Regarding net basin supply (NBS), "other" supplies, and "other" diversions (see Section 4.3 for further information); positive values indicate flow into a lake, while negative values indicate flow out of a lake. Consumptive use flows, the Long Lac/Ogoki Diversions, the Chicago Diversion, and Welland Canal flow are always positive.

One record near the top of the file will give the units that data is expressed in, beginning with the string "UNITS:" followed by one of:

- m3s cubic meters per second (flow)
- 10m3s tens of cubic meters per second
- cfs cubic feet per second
- tcfs thousands of cubic feet per second
- inches inches
- feet feet
- cm centimeters
- mm millimeters
- m meters (distance or elevation)
- m2 square meters (area)
- ft2 square feet
- mi2 square miles
- km2 square kilometers

- m3 cubic meters (volume)
- ft3 cubic feet

The units m3s, m, m2, and m3 are designated as the basic units for flow, distance, area, and volume. All data read in is immediately converted to one of these units, before being used. All data lines conform to a single, fixed-column format. For monthly data, each record lists the year, followed by twelve monthly values in fields of 8-columns each. Quarter-monthly data records list the year and quarter-month, followed by 12 values (one for each month) applicable to that quarter of the month, also in 8-column fields (see example following). Daily data records list the year, month, and quarter-month, followed by 7 or 8 values (one for each day) applicable to that quarter of the month, also in 8-column fields. The weekly data format consists of the date (year, month, and day) corresponding to the end of the week, followed by a single 8-column value. For Lake Ontario regulation, weeks always end on a Friday.

```
# LAKE ERIE QUARTER MONTHLY OUTFLOWS (10m^3/s)
# Comment field
# 1900    1991
# Fortran format: I4, 12F8.0
INTERVAL: Quarter-Monthly
UNITS: 10m3s
# Yr QM    J      F      M      A      M      J      J      A      S      O      N      D
1900 1   547.   518.   544.   555.   583.   592.   583.   578.   572.   536.   569.   558.
1900 2   527.   532.   538.   555.   592.   589.   583.   578.   572.   541.   580.   544.
1900 3   530.   524.   555.   561.   586.   578.   575.   569.   566.   544.   561.   541.
1900 4   558.   544.   544.   569.   583.   595.   572.   572.   544.   535.   527.   569.
1901 1   527.   504.   481.   496.   513.   544.   555.   538.   541.   530.   532.   510.
…
```

Another record near the top of the file shows the time interval of the data (e.g., "Quarter-Monthly"), though only the first letter is significant, and case doesn't matter (can be m, q, w, or d; M, Q, W, or D).

Any line beginning with a pound character ("#") is a comment. In general, output files should provide at least this much documentation near the top:

4) type of data (e.g., "net basin supplies")

5) time period contained in file (e.g., "(January) 1900 to (December) 1990")

6) Fortran format statement (e.g., "I4, 12F8.0"). Here, this example tells the user to expect each line of data to contain four-character integer, followed by a dozen real numbers, each with eight characters, and zero decimal places.

Attempts have been made to ensure that these files adhere to a strict naming convention. The first two characters refer to the location of the data (i.e., "sp" for Lake Superior or for "nr" for Niagara River). The third character indicates the time increment of the data ("m" for monthly, "q" for quarter-monthly, "w" for weekly, "d" for daily, or a hyphen ("-") for mixed time periods or other unusual arrangements). The fourth, fifth, and sixth characters describe the type of data (e.g., "nbs" for net basin supplies, or “div" for diversion). The seventh and eighth characters in the filename are optional and not normally used. An optional extension of up to six characters may also appear. See “Output Extension” in Section 4.3.1 for full details. For example, mean monthly forecasted outflows for the St. Clair River stemming from a Lake Superior regulation run to determine December's outflow may reside in a file named crmflo.dec50. A comprehensive list of allowable data file names is given near the end of Section 4.1. The program also uses this convention when writing optional output files. Currently, the CGLRRM does not support filenames that include spaces.

6.5 Error and Warning Messages

All error and warning messages lie in a file named "MESSBASE.TXT". This is the error/warning message database that contains the text blocks the program uses to convey messages to the user whenever an error or other potential problem is discovered during model processing. Currently, there are approximately 175 messages contained in the database. The full contents of this file are included in Appendix C.

6.6 Debugging

Debugging refers to the process of finding and fixing all errors in a model. Though the CGLRRM has been designed to be extremely user-friendly, it is imperative that input files be constructed carefully to preclude unwanted errors in model results. There are two types of errors that can exist in input files: syntax errors and logic errors. When a parameter input code or data is entered incorrectly, a syntax error is created. When the user enters "Other Supplies" for Lake Erie that already exist in the net basin supplies also given, a logic error is created. The model may find many syntax and logic errors during processing, but it will always be up to the user to ensure that input files are accurate and are designed to permit the model to perform the desired task effectively.

Many features have been included in the model design to help users during the debugging process. It is highly recommended that users employ high values for "Verbosity" settings until they are comfortable with the model and their results. Users should take full advantage of the "DETAILED.LOG" and "FAILURE.LOG" files discussed in Section 6.3 during debugging. Of course, whenever an error or warning message is received, users should pay close attention to the message(s), and endeavor to fix each portion of input to preclude further messages, wherever possible. See Appendix C for a complete listing of error and warning messages.

7 CASE STUDIES

7.1 Lake Superior Regulation

For this case study, users are referred to directory "…\SUPREG\…". For further information regarding specific input parameters, please refer to Section 4.3.

Traditional Lake Superior regulation practice uses three six-month simulations of different supply scenarios, although the regulation decision is based entirely on results from the first month of one of the scenarios. This directory contains three runs to show how to make traditional Lake Superior Regulation Plan 1977-A model runs using the CGLRRM. Input ("dmpl77a") and output ("ompl77a") for the USACE model are provided for comparison.

As with all CGLRRM runs, the input starts with the file "cglrrm.ini". In this case, it points to three parameter files. The number of parameter files used in any particular run is up to the modeler. Since this sample involves a simulation that would likely be repeated every month (each time varying only a few of the parameters) the three-file arrangement provides a degree of clarity and safety.

The first parameter file is "../BIN/SYSTEM.DAT", which contains information specific to a given installation of the model. The only parameters listed in "../bin/system.dat" are:

OnlyOneOpen:	False
Message Database:	...\bin\messbase.txt
Modeler Name:	(whatever was typed during install)

OnlyOneOpen is rarely needed, and could be omitted on most modern systems. The location of the message database is not absolutely required to run the model, but it greatly increases the feedback that the model can provide in case of problems, and is highly recommended. The modeler name is also optional. In this case, it actually gets overridden in a subsequent parameter file. The point of putting these three parameters in "system.dat" is that they are typically the same for all the runs done at a given machine, and by listing "system.dat" at the top of the cglrrm.ini files, they needlessly lengthen other parameter files.

The second parameter file listed in cglrrm.ini is "..\bin\supreg.dat". This file contains all the parameters regarding Lake Superior regulation runs that do not change. None of the parameters (except General Verbosity) in this file may be redefined in a subsequent parameter file (even if the exact same values are used). The first parameter specifies that the run includes all of the upper lakes. The second parameter specifies that storage change computations should be based on "uniform" months, as required by traditional practice. The next parameters specify rounding practices that match those of the previous models.

The next parameters apply to Lake Superior. The first of them specifies the traditional monthly routing interval. The Sup Inc parameter conforms to the traditional practice of not subdividing each month into multiple increments when routing, so that the Superior mean and EOM levels will be calculated assuming that outflow is constant during the month. The next parameter specifies the (constant) area of Lake Superior. The area given was determined from the 31,380 ($m^3/s\forall$months)/m constant, in order to be compatible with a uniform month. The next parameter directs the model to use Plan 1977-A to determine St. Mary’s River outflows (as opposed to an outflow equation). This necessitates the next three parameters (P77A Compat Mode, Sup Criteria Round, Sup S.W. Pier Formula; see Section 4.3.2). The final parameter in the Superior section specifies the Long Lac/Ogoki diversion values used for regulation purposes.

The next set of parameters applies to the middle lakes. The first one is a title line, followed by a parameter specifying that routing should be on a quarter-monthly basis. The next parameter specifies that the storage change in each middle lake should be integrated over 10 increments per quarter-month. The next parameter indicates that the nine simultaneous routing equations for each increment should be solved using an iterative procedure, as opposed to GLERL’s matrix solution. The next three parameters provide area for the middle lakes, determined similarly as for Lake Superior. The next three parameters describe the outflow equations used for the middle lakes. The next two parameters quantify the middle lake diversions amounts used for Superior regulation. The final three parameters specify the middle lake outflow retardations traditionally used for Superior regulation.

The third parameter file listed in cglrrm.ini is "2000oct50.dat". This file contains all the remaining parameters needed for the run evaluating the 50% supply scenario for October 2000. The first two are job titles. The next two specify the duration of the run (a six-month forecast). The next four parameters provide the levels of the lakes at the start date. In this example, the beginning of the routing period (i.e., month or quarter-month) coincides with the start date. If the start date was given as October 3, the levels for October 3 should be specified. The model adjusts the durations of the first and last routing periods in order to accommodate the mixing of daily, weekly, quarter-monthly, and monthly intervals. Note that Plan 1977-A requires beginning-of-month levels. In this example, the start levels are actually beginning-of-month levels, so no additional input is required. If the start date had been given as October 3, then the parameters SUP BOM LEVEL, MHU BOM LEVEL, etc., would be necessary. Plan 1977A also needs to know the previous month's output (i.e., last month's St. Mary's River outflow), which is given in the line after the start levels.

The next two parameters specify a directory to write output files to, and the filename extension to use. The next parameter overrides the modeler name specified in "system.dat". If this sample is copied and modified for actual use, this line should be deleted.

The next four parameters are the supplies to use for Superior Criterion evaluations and routing after the regulation decision. Note that the model's application of these supplies is contingent on the P77ACompat parameter specified earlier (see Section 4.3.2). In this example, the CGLRRM is using the same logic as previous models, and performs the criterion checks and sidechannel calculations using these supplies. In order to replicate existing regulation practice, the supplies used in these parameters must be drawn from this "operational" supply matrix. Each of the twelve sequences for each lake is included here. The modeler must simply uncomment the appropriate block corresponding to the starting date's month used in the run.

The next three parameters direct the model to write out monthly averages for the levels and flows on the middle lakes. The standard report contains only results for each routing period (quarter-month), so if averages of the routing results are needed, they must be requested. This output gets written in the standard format, in files named according to the convention described in Section 6.4.

The final parameters adjust the degree of output that will be written to the detailed log file.

The run is executed by opening up a command-line window and changing to the directory containing the cglrrm.ini file (in this example, "supreg"). If it does not already exist, create the user-specified output directory (the model does not create directories). Run the model by typing its name (i.e., "...\bin\cglrrm").

Messages will appear on the console listing each parameter file as they are read, and will mention anything unusual or suspicious about the data being parsed. The amount of this feedback is controlled by the value of GeneralVerbosity in effect at the time the line is parsed. In this example, the model indicates that it did a unit conversion on an area specified in "supreg.dat".

After the parameter files are read, the model enters an input-checking module, looking for consistency, completeness, and redundancy. Here, many common mistakes are often identified. If users find themselves making input errors that the checking module fails to catch, they are asked to describe the error to the model custodian so that it can be added to the list of checks to make. Most of these conditions abort the run, but some result in warnings. In addition to controlling the output written to the detailed log file, the CheckVerbosity parameter also controls whether warnings are written to the console. In this case, the checking module warns that we are routing through a lake (Superior) without specifying any retardation losses. This is the traditional practice, so discount this warning.

After the checking module, the model preprocesses all of the time-series data so that it is extracted and adjusted to the routing intervals appropriate to its lake. This results in the files "spmtsd.oct50", "mhqtsd.oct50", "scqtsd.oct50", and "erqtsd.oct50" in the output directory. The format of these files is described in Section 6.3. Depending on the settings of TSDataVerbosity, the intermediate calculations for each line in these files may be written as comments between the lines.

After the time-series data files are established, the model begins routing/regulation computations. The time-series data files are opened and read one line at a time (one for each routing period). Information regarding the computations is written to "detailed.log". The amount of information is controlled by the variables SupVerbosity, MidlakeVerbosity, and GeneralVerbosity. Modelers are encouraged to read through the messages in order to follow the logic flow. It may be necessary to look up certain messages in the source code where they originate to fully understand them. The volume of data can be quite significant (at maximum settings it can exceed 1 MB/year of simulation). Depending on these settings, unusual or suspicious conditions that arise during the routing may prompt warning messages.

In this example, the model reported that the routing solution for the one increment per month on Lake Superior failed the continuity test five (out of six) times. The maximum error was 120 m^3s. This is due to rounding the levels to the nearest centimeter. Reviewing "detailed.log" shows that the one time that the Lake Superior system balanced to within 1 m^3s was Jan 2001. The error of the routing solutions on the middle lakes exceeded the 1 m^3s warning threshold 240 times (i.e., every increment). The worst case was 89 m^3s. Which lake this happened on and when it occurred can be found in "detailed.log". In this case, rounding levels to four places (i.e., one-tenth of a millimeter) causes the error. In both instances, these errors are accepted as part of traditional practice, and the messages can be ignored.

At the conclusion of the run, the model wrote the summary file to the output directory.

Also, as directed by "MHu Output Files" and related parameters, monthly mean levels and flows for each middle lake were written to the output directory, in files crmflo.oct50, drmflo.oct50, ermmlv.oct50, mhmmlv.oct50, nrmflo.oct50, and scmmlv.oct50

The output directory also contains reports named "micherie.oct50" and "superior.oct50". "micherie.oct50" tabulates the levels and flows computed on each middle lake for each routing interval. “superior.oct50" does the same for Lake Superior, but since Plan 1977-A was used, it contains additional information (Section 6.3).

The last file to be described is the one named "failure.log". This accumulates the warning and error messages of all the runs from this directory.

This completes the demonstration for the first of the three traditional regulation runs. The other two are very similar. For the second run, the user needs to edit “cglrrm.ini” and replace the line containing "2000oct50.dat" with "2000Oct05.dat". Inside the "2000Oct05.dat" file, notice that the supplies are different, reflecting the alternate scenario. Moreover, currently only the supplies for October are present in this file (and 2000 Oct95.dat). Modelers presently need to enter the supplies for the other months, as needed.

The only other changes to make are various labeling changes regarding the titles, output directory, and extension. Run the model again.

Repeat this process yet again for the third supply scenario, substituting "2000Oct95.dat" for "2000Oct05.dat" in "cglrrm.ini".

This completes the demonstration of all three traditional Lake Superior Regulation Plan 1977-A runs.

7.2 6-Month Great Lakes Forecast

This case study has yet to be prepared.

7.3 Lake Ontario Regulation

This case study has yet to be prepared.

7.4 Simulation

This case study has yet to be prepared.

8 DEVELOPMENTAL ISSUES

8.1 Outstanding Items

Since development of the CGLRRM is a long-term, open-ended project, there are some major phases of development yet to be addressed or completed. The following list highlights the most significant items. The first two items are considered highest priority at the moment. Most of these items are highly technical in nature, and have been included here solely for interest’s sake.

1) The Ontario/St. Lawrence module's place in the overall architecture is established, and much of the lower lake's input and output is functioning, but the Plan 1958-D module is

very new (incorporated July 2000). Incorporating Plan 1958-D is complicated by the desire to preserve the flexibility and lack of hardcoded assumptions that exists elsewhere in the CGLRRM. Plan 1958-D has a lot more operational gray area and interdependent assumptions. Moreover, any tiny alteration to existing practice, no matter how logical the change or better the engineering, is expected to be perceived as a detriment to some interest, and require a major administrative and political effort. Testing is expected to take some time.

2) The major phase yet to be initiated is development of a graphical user interface (GUI), serving as both a front-end for preparing inputs, and as a post-processor for creating attractive reports and graphics. Design objectives for this work are currently being finalized.

3) Determination of when to give output for a period partially routed. When routing begins partway through a period, or finishes before the end of one, should the average be calculated (e.g., if using weekly routing starting on 12th of month, should there be a monthly average for the first month?).

4) Revise calculation of supplies, etc., for runs that begin or end partly through a routing period. When routing begins after the start of a period, or finishes before the end of one, how should the period-average NBS be calculated if using input data of a shorter period (e.g., monthly simulation begins Jan. 9, 1982 (start of quarter 2). Given quarter-monthly NBS, should the January NBS in the routing computation use the average of all four quarter-monthly NBS's, or just the three that the simulation period includes)? Currently, all four quarter-months are used, since that is existing practice for programs using separate files for monthly or quarter-monthly data versions. However, it is best to use the data that most closely matches the simulation period.

5) Implement routing of probabilistic supplies to permit reading in a probability density curve for NBS instead of specific values. When routing computations call for a supply, a random number generator reads one from the curve, and it is routed. Repeat to yield level and flow frequency curves for each routing interval. Problems include how to handle auto-correlation of supplies, and supplies that partially-correlate to neighboring basins. Implementation would require significant effort, due to output complexity.

6) Recoding is necessary to ensure that NTS is computed in same loop as routing calculations, instead of beforehand. There should be a noticeable performance gain and it may be preferable while debugging output. Considerable effort will be required, however.

7) Support embedded spaces in filenames. This requires modifying string/filename-parsing routines to recognize filenames with embedded spaces, and verifying that file open/close routines and associated messages support such names.

8) Check that all the data required to run Plan 1977-A have been read. Another check would need to be included in the input parameter checking subroutine if the design changes to eliminate regulation plan information from the model. Currently all Plan

1977-A data is included as defaults, with many values implemented as inputs that may optionally override the defaults.

9) Code in post-regulation decision evaluations of Criteria (b) and (c) for Lake Superior regulation. Indicate how much different the flow would have had to be in order to have met the criteria. Existing practice cannot predict what Plan 1977-A or criteria outflows from Lake Superior will be, except in the first month of simulation. The "observed" supply for the upcoming month is used when computing the gate setting closest to the plan flow, or when computing the criteria flows. Obviously, such supplies are not available when deciding the outflow for the month, so simulations are flawed. "Operational" supplies used can be incorporated into the model (already are for Criterion (b)), and used to replicate the regulation decision independent of the "observed" NBS. But after the period ends, one could check how close the operational NBS was to the observed, and if the regulated flow was in violation of the criterion.

10) Enable forced gates or forced outflow from Lake Superior. This would require addition of a forced flow times-series input parameter, and adding a column for it to the summary input file. This already exists for gates. Logic coding would be required as well to test for forced flow or forced gates, and (if specified) short-circuit the regulation plan or outflow equation. This is already partially done. Forced flows are implemented, but which gate equation to use for forced gates must still be decided.

11) Find out about whether or not there are defaults for subscripts on CHARACTER variables. Currently it is assumed that default subscripts are part of the ANSI standard for Fortran 77. If the number to the left of ':' is absent, the compiler uses 1. If the number to the right of ':' is absent, the compiler uses the dimensioned length of the string. Some older compilers require explicit limits. 1's can be added to all the empty left sides of the ':', and LEN(stringname) to all the empty right sides in this case.

12) Make the Point Iroquois/Southwest Pier conversion, and gate constants readable inputs. Currently this is hardcoded, but this is unacceptable.

13) The program logic must be illustrated in a flow chart to be included in this manual.

14) An organized input description will be made available in future versions.

15) Additional parameter checks will be added to see if unnecessary parameters were specified: NBS, diversions, equations, retardations, consumptive uses, start levels, plan data, etc.

8.2 Known Bugs

Since development of the CGLRRM is a long-term, open-ended project, there are some minor outstanding programming bugs yet to be addressed. The following list highlights the bugs that shall be addressed in future versions. Users are urged to let the model custodian know of any bugs uncovered during model use and debugging which are not included in the following list.

1) For parameters "SUP EOP LEVEL DATA", "MHU EOPLEVEL DATA", "STC EOP LEVEL DATA", "ERI EOP LEVEL DATA", and "ONT EOP LEVEL DATA" (see Section 4.3.5), the program currently converts to meters before looking to see if the value is –9999, and fails to recognize missing data. Until fixed, units of “m” must be used. No more than two decimal positions will be used.

2) For parameters "SUP OUTFLOW DATA" and “SUP SIDECHAN DATA” (see Section 4.3.2), the program currently converts to cubic meters per second before looking to see if the value is –9999, and fails to recognize missing data. Until fixed, units of “m3s” must be used. No more than one decimal position will be used.

3) When weekly routing is chosen for Lake Superior under Plan 1977-A, the weighted averages for weeks that overlap between months are improperly computed. Currently, the previous month’s inflow is assumed to be that from the previous week. Also, the routing calculations for weeks straddling the end of months fail to take a weighted average of the number of gates and the non-gated flow in effect for the week (e.g., if the last day of the week is the first of the month, then the new month’s gate setting and non-gated flow are applied to the entire week). Until fixed, users should only consider weekly results for those weeks falling completely within months for which Plan 1977-A is invoked.

8.3 Troubleshooting

Since the CGLRRM is a new model, still in development, bugs will undoubtedly arise. Users are urged to contact the model custodian with any problems so that a comprehensive troubleshooting table can be prepared.

Currently, the model custodian is Mr. Roger Gauthier, Head of Hydrology, USACE Detroit District:

Mail: 477 Michigan Avenue
Detroit MI 48226
Phone: (313) 226-3054
E-mail: Roger.L.Gauthier@lre02.usace.army.mil

8.4 FAQs

The following list contains some anticipated questions that most users may find helpful.

1) Since the CGLRRM is under development, where can I find the latest version easily?

Efforts are made to keep the latest documentation and installation package available on the CGLRRM Web site:

http://huron.lre.usace.mil/cglrrm/

2) Is it necessary to register to obtain a copy of the model?

A binational panel of government experts from the U.S. and Canada developed the CGLRRM. It is available as public-domain freeware, and you need not register to obtain a copy. However, users are urged to register via the model custodian in order that timely updates may be made available from time-to-time. A list of registered users will also help illustrate which users are most interested in the product as it is developed. Those people who chose to register are urged to offer any comments or suggestions at that time as well.

3) *I'm not a water resources expert. Can I use the model?*

Although water management experts as well as the public designed the CGLRRM for use, users need some knowledge of hydrologic and hydraulic routing principles and/or current regulatory practices to perform model runs effectively, as well as comprehend the technical output.

4) *I don't like juggling around of ASCII files. When will the graphical user interface (GUI) be ready?*

Currently, the front-end (i.e., input) GUI is in the early design stage. It is not anticipated to be complete for a few years.

A GUI for use of the CGLRRM as part of GLERL's AHPS package is expected later this year. A GUI for Lake Superior regulation should be available as early as this Summer.

5) *I would like to perform some statistical analyses on the results and produce some comparative plots. Can I somehow do this directly in the model settings?*

There is no capability to perform any post-analysis or plotting directly within the model at present. Users must currently process output data using other software, as desired. A back-end (i.e., output) graphical user interface (GUI) may be incorporated in the future which will provide some simple statistical and plotting features. Users are urged to offer suggestions with regard to this potential addition.

If questions remain after checking through this manual, users are urged to contact the model custodian so that a comprehensive list of "frequently-asked questions" can be prepared.

8.5 Reporting Problems

Since the CGLRRM is a new model, undergoing dynamic development issues at the present time, inherent bugs and problems will undoubtedly arise. Users are urged to contact the model custodian with any problems and concerns so that they can be addressed in a timely manner, and corrected for future versions.

Currently, the model custodian is Mr. Roger Gauthier, Head of Hydrology, USACE Detroit District. Mr. Gauthier can be reached at:

Mail: 477 Michigan Avenue
Detroit MI 48226
Phone: (313) 226-3054
E-mail:Roger.L.Gauthier@lre02.usace.army.mil

Any errors, omissions, or suggestions with regard to this User's Manual may be addressed to Mr. Gauthier, or the head author, Mr. Rob Caldwell, Water Resources Engineer, Environment Canada. Mr. Caldwell can be reached at:

Mail: 111 Water Street East
Cornwall ON K6H 6S2
Phone: (613) 938-5725
E-mail:Rob_Caldwell@pch.gc.ca

8.6 Getting Help

Since the CGLRRM is undergoing dynamic development issues at the present time, direct help lines have yet to be established. Users in need of advanced assistance are urged to contact the model custodian. He will endeavor to answer all queries quickly or direct users to the best source of available information.

Currently, the model custodian is Mr. Roger Gauthier, Head of Hydrology, USACE Detroit District. Mr. Gauthier can be reached at:

Mail: 477 Michigan Avenue
Detroit MI 48226
Phone: (313) 226-3054
E-mail:Roger.L.Gauthier@lre02.usace.army.mil

Appendix A

INPUT PARAMETERS

QUICK REFERENCE

The following ***alphabetic*** reference list is meant to serve as a quick cross-reference for users. For complete details on each parameter, please *refer to Section 4.3*. Note that the *item numbers* (e.g., 1-1) match those listed in Section 4.3. The first number is the sub-section (e.g., 4.3.1), and the second number is the (arbitrary) parameter number within this sub-section.

BOM ROUND **1-16**
Determines number of decimal places left after rounding of BOM values, expressed as powers of ten (optional).

CGIP AVG LEVEL DATA **3-7**
Stage to use for the Niagara River's Chippewa-Grass Island Pool, which serves as the downstream level in two-gage discharge relationships for the river (normally 171.16 m).

CGIP START LEVEL **3-8**
Stage to use for the Niagara River's Chippewa-Grass Island Pool, which serves as the downstream level in two-gage discharge relationships for the river (normally 171.16 m).

CHECK VERBOSITY **1-4**
Controls the degree of debugging information written to the screen or the detailed log file during the check routine.

CHICAGO DIV DATA **5-44**
Chicago diversion discharge from L. Michigan.

COMB LLO DIV DATA **5-43**
Sum of the Long Lac and Ogoki diversions into Lake Superior.

DET RIV EQN **3-5**
Discharge equation parameters for the Detroit R.: equation coefficient(K), channel invert (YM), depth exponent (A), fall exponent (B), downstream lake weighting factor (Wt), and fall adjustment constant (Zc) where
$Q=K(Z_u*Wt+(1-Wt)Z_d-YM)^A(Z_u-Z_d+Z_c)^B$

DET RIV FLOW DATA **5-55**
Flow in the Detroit R.

DET RIV ICE DATA **5-63**
Ice retardation in the Detroit R.

DET RIV RETR DATA **5-65**
Combined ice and weed retardation in the Detroit R.

DET RIV WEED DATA **5-64**
Weed retardation in the Detroit R.

DPR RIV FLOW DATA **4-2**
The amount of Des Prairies R. flow.

END DATE **1-12**
End date of simulation (required).

ERI AREA-ELEV **5-6**
Area vs. elevation data for L. Erie (maximum 5 points).

ERI BOM LEVEL **2-19**
Beginning-of-month level for L. Erie (required for Plan 1977-A, if "START DATE" is not the first of a month).

ERI CON USE DATA **5-32**
Consumptive use for L. Erie.

ERI EOP LEVEL DATA **5-75**
Forced end-of-period level for L. Erie. See Section 8.2 for bug information.

ERI NBS DATA **5-25**
Net basin supply data for L. Erie.

ERI OTHER DIV DATA **5-51**
Other diversions for L. Erie.

ERI OTHER SUP DATA **5-39**
Other supplies for L. Erie.

ERI OUTPUT FILES **1-23**
Indicates which output files are desired for L. Erie, in addition to the standard results file (optional).

ERI START LEVEL **5-11**
Initial lake level for L. Erie (required if routing through the lake).

ERI SUM INPUT FILE **5-80**
Summary file from which to read time-series input data for L. Erie.

FLOW ROUND **1-18**
Determines number of decimal places left after rounding of outflow values, expressed as powers of ten (optional).

GENERAL VERBOSITY **1-2**
Controls the degree of debugging information written to the screen or the detailed log file.

HUR AREA-ELEV **5-4**
Area vs. elevation data for L. Huron (maximum 5 points).

HUR BOM LEVEL **2-17**
Beginning-of-month level for L. Huron

HUR CON USE DATA **5-30**
Consumptive use for L. Huron.

HUR NBS DATA **5-23**
Net basin supply data for L. Huron.

HUR OTHER DIV DATA **5-49**
Other diversions for L. Huron.

HUR OTHER SUP DATA **5-37**
Other supplies for L. Huron.

LAKES TO SOLVE FOR **1-1**
Which lake levels to solve for (required).

LEVEL ROUND **1-17**
Determines number of decimal places left after rounding of water levels, expressed as powers of ten (optional).

LONG LAC DIV DATA **5-41**
Long Lac diversions into Lake Superior.

LS ROUGHNESS DATA **4-5**
Coefficients for Long Sault roughness data.

MESSAGE DATABASE **1-9**
Filename location of message database (optional).

MHU AREA-ELEV **5-2**
Area vs. elevation data for L. Michigan-Huron (maximum 5 points).

MHU BOM LEVEL **2-15**
Beginning-of-month level for L. Michigan-Huron (required for Plan 1977-A, if "START DATE" is not the first of a month).

MHU CON USE DATA **5-28**
Consumptive use for L. Michigan-Huron.

MHU EOP LEVEL DATA **5-73**
Forced end-of-period level for L. Michigan-Huron. See Section 8.2 for bug information.

MHU NBS DATA **5-21**
Net basin supply data for L. Michigan-Huron.

MHU OTHER DIV DATA **5-47**
Other diversions for L. Michigan-Huron.

MHU OTHER SUP DATA **5-35**
Other supplies for L. Michigan-Huron.

MHU OUTPUT FILES **1-21**
Indicates which output files are desired for L. Michigan-Huron, in addition to the standard results file (optional).

MHU START LEVEL **5-9**
Initial lake level for L. Michigan-Huron (required if routing through the lake).

MHU SUM INPUT FILE **5-78**
Summary file from which to read time-series input data for L. Michigan-Huron.

MIC AREA-ELEV **5-3**
Area vs. elevation data for L. Michigan (maximum 5 points).

MIC BOM LEVEL **2-16**
Beginning-of-month level for L. Michigan.

MIC CON USE DATA **5-29**
Consumptive use for L. Michigan.

MIC NBS DATA **5-22**
Net basin supply data for L. Michigan.

MIC OTHER DIV DATA **5-48**
Other diversions for L. Michigan.

MIC OTHER SUP DATA **5-36**
Other supplies for L. Michigan.

MIDLAKE INC **5-16**
Number of sub-increments within each routing interval for middle lakes. It is recommended to use increments of 222,000 s or less (about 2.5 d).

MIDLAKE VERBOSITY **1-6**
Controls the degree of debugging information written to the screen or the detailed log file during middle lakes routing.

MIDLAKE SOLUTION **3-1**
Determines whether to use the iterative or matrix solution to solve nine simultaneous equations for the middle lakes.

MIDLAKE TIMESTEP **5-14**
Routing interval to use for L. Michigan-Erie (required if routing through these lakes).

MII RIV FLOW DATA **4-3**
The amount of Milles Iles R. flow.

MONTH LENGTH **1-19**
Either uniform (traditional practice) or actual (optional).

NIA RIV EQN **3-6**
Discharge equation parameters for the Niagara R.: equation coefficient(K), channel invert (YM), depth exponent (A), fall exponent (B), downstream lake weighting factor (Wt), and fall adjustment constant (Zc) where
$Q=K(Z_u*Wt+(1-Wt)Z_d-YM)^A(Z_u-Z_d+Z_c)^B$

NIA RIV FLOW DATA **5-56**
Flow in the Niagara R.

NIA RIV ICE DATA **5-66**
Ice retardation in the Niagara R.

NIA RIV RETR DATA **5-68**
Combined ice and weed retardation in the Niagara R.

NIA RIV WEED DATA **5-67**
Weed retardation in the Niagara R.

OGOKI DIV DATA **5-42**
Ogoki diversions into Lake Superior.

ONT AREA-ELEV **5-7**
Area vs. elevation data for L. Ontario (maximum 5 points).

ONT CON USE DATA **5-33**
Consumptive use for L. Ontario.

ONT DEVIATION DATA **4-4**
The amounts of any deviations from Lake Ontario regulation Plan 1958-D.

ONT EOP LEVEL DATA **5-76**
Forced end-of-period level for L. Ontario. See Section 8.2 for bug information.

ONT INC **5-19**
Number of sub-increments within each routing interval for L. Ontario. It is recommended to use increments of 222,000 s or less (about 2.5 d).

ONT NBS DATA **5-26**
Net basin supply data for L. Ontario.

ONT OTHER DIV DATA **5-52**
Other diversions for L. Ontario.

ONT OTHER SUP DATA **5-40**
Other supplies for L. Ontario.

ONT OUTPUT FILES **1-24**
Indicates which output files are desired for L. Ontario, in addition to the standard results file (optional).

ONT START LEVEL **5-12**
Initial lake level for L. Ontario (required if routing through the lake).

ONT SUM INPUT FILE **5-81**
Summary file from which to read time-series input data for L. Ontario.

ONT TIMESTEP **5-15**
Routing interval to use for L. Ontario (required if routing through the lake).

ONT VERBOSITY **1-7**
Controls the degree of debugging information written to the screen or the detailed log file during L. Ontario routing.

OUTPUT DIRECTORY **1-10**
Directory location of output files (optional).

OUTPUT EXTENSION **1-8**
Extension to use for output files (optional).

P77A COMPAT MODE **2-2**
Describes which supplies, diversions, and retardations to use between determining Plan 1977-A's five months of forecasted outflows, and determining the actual gate setting and allotted sidechannel flow for the upcoming month.

P77A DET RIV EQN **2-8**
Discharge equation parameters to use with Plan1977-A for the Detroit R. (optional). Includes equation coefficient(K), channel invert (YM), depth exponent (A), fall exponent (B), downstream lake weighting factor (Wt), and fall adjustment constant (Zc) where $Q=K(Z_u*Wt+(1-Wt)Z_d-YM)^A(Z_u-Z_d+Z_c)^B$

P77A FLOW ROUND **2-10**
Rounding parameter for Plan 1977-A outflows, expressed as powers of ten (optional).

P77A LEVEL ROUND **2-11**

Rounding parameter for Plan 1977-A water levels, expressed as powers of ten (optional).

P77A MIDLAKE INC **2-5, 5-17**

Positive integer number of sub-increments within each monthly routing interval for the middle lakes routing built into Plan 1977-A (optional). It is recommended to use increments of 222,000 s or less (about 2.5 d.).

P77A MONTH ROUND **2-12**

Rounding parameter for months within Plan 1977-A, expressed as powers of ten (optional).

P77A NIA RIV EQN **2-9**

Discharge equation parameters to use with Plan1977-A for the Niagara R. (optional). Includes equation coefficient(K), channel invert (YM), depth exponent (A), fall exponent (B), downstream lake weighting factor (Wt), and fall adjustment constant (Zc) where $Q=K(Z_u*Wt+(1-Wt)Z_d-YM)^A(Z_u-Z_d+Z_c)^B$

P77A STC RIV EQN **2-7**

Discharge equation parameters to use with Plan1977-A for the St. Clair R. (optional). Includes equation coefficient(K), channel invert (YM), depth exponent (A), fall exponent (B), downstream lake weighting factor (Wt), and fall adjustment constant (Zc) where $Q=K(Z_u*Wt+(1-Wt)Z_d-YM)^A(Z_u-Z_d+Z_c)^B$

P77A STM RIV EQN **2-6**

Discharge equation parameters to use with Plan1977-A for the St. Mary's R. (optional). Includes equation coefficient(K), channel invert (YM), depth exponent (A), fall exponent (B), downstream lake weighting factor (Wt), and fall adjustment constant (Zc) where $Q=K(Z_u*Wt+(1-Wt)Z_d-YM)^A(Z_u-Z_d+Z_c)^B$

PREV STM RIV FLOW **2-13**

The previous month's Lake Superior outflow (required for Plan 1977-A).

ROUTING TITLE **3-2**

Description applicable to middle lakes routing (optional).

START DATE **1-11**

Start date of simulation (required).

STC AREA-ELEV **5-5**

Area vs. elevation data for L. St. Clair (maximum 5 points).

STC BOM LEVEL **2-18**

Beginning-of-month level for L. St. Clair (required for Plan 1977-A, if "START DATE" is not the first of a month).

STC CON USE DATA **5-31**
Consumptive use for L. St. Clair.

STC EOP LEVEL DATA **5-74**
Forced end-of-period level for L. St. Clair. See Section 8.2 for bug information.

STC NBS DATA **5-24**
Net basin supply data for L. St. Clair.

STC OTHER DIV DATA **5-50**
Other diversions for L. St. Clair.

STC OTHER SUP DATA **5-38**
Other supplies for L. St. Clair.

STC OUTPUT FILES **1-22**
Indicates which output files are desired for L. St. Clair, in addition to the standard results file (optional).

STC RIV EQN **3-4**
Discharge equation parameters for the St. Clair R.: equation coefficient(K), channel invert (YM), depth exponent (A), fall exponent (B), downstream lake weighting factor (Wt), and fall adjustment constant (Zc) where
$Q=K(Z_u*Wt+(1-Wt)Z_d-YM)^A(Z_u-Z_d+Z_c)^B$

STC RIV FLOW DATA **5-54**
Flow in the St. Clair R.

STC RIV ICE DATA **5-60**
Ice retardation in the St. Clair R.

STC RIV RETR DATA **5-62**
Combined ice and weed retardation in the St. Clair R.

STC RIV WEED DATA **5-61**
Weed retardation in the St. Clair R.

STC START LEVEL **5-10**
Initial lake level for L. St. Clair (required if routing through the lake).

STC SUM INPUT FILE **5-79**
Summary file from which to read time-series input data for L. St. Clair.

STL RIV ICE DATA **5-69**
Ice retardation in the St. Lawrence R.

SUP EOP LEVEL DATA **5-72**
Forced end-of-period level for L. Superior. See Section 8.2 for bug information.

SUP GATES DATA **2-21**
The number of gates used for forced gate settings.

SUP INC **5-18**
Number of sub-increments within each routing interval for L. Superior. It is recommended to use increments of 222,000 s or less (about 2.5 d).

SUP NBS DATA **5-20**
Net basin supply data for L. Superior.

SUP OTHER DIV DATA **5-46**
Other diversions for L. Superior.

SUP OTHER SUP DATA **5-34**
Other supplies for L. Superior.

SUP OUTFLOW DATA **2-22**
The total St. Mary's River flow used for forced flow settings. See Section 8.2 for bug information.

SUP OUTPUT FILES **1-20**
Indicates which output files are desired for L. Superior, in addition to the standard results file (optional).

SUP SIDECHAN DATA **2-23**
The total non-gated flow for the St. Mary's River, which should be specified if forced gate settings are used (otherwise, the maximum sidechannel flow (2320 m3s) is assumed). See Section 8.2 for bug information.

SUP START LEVEL **5-8**
Initial lake level for L. Superior (required if routing through the lake).

SUP S.W. PIER FORMULA **2-4**
Indicates whether to use the old relationship for converting Lake Superior levels to S.W. Pier levels (1), or the new Environment Canada (2000) equation (2).

SUP SUM INPUT FILE **5-77**
Summary file from which to read time-series input data for L. Superior.

SUP TIMESTEP **5-13**
Routing interval to use for L. Superior (required if routing through the lake).

SUP VERBOSITY **1-5**
Controls the degree of debugging information written to the screen or the detailed log file during L. Superior routing.

TITLE 1 **1-13**
First line of descriptive information applicable to the run (optional).

TITLE 2 **1-14**
Second line of descriptive information applicable to the run (optional).

TITLE 3 **1-15**
Third line of descriptive information applicable to the run (optional).

TS DATA VERBOSITY **1-3**
Controls the degree of debugging information written to the time series data files.

U STL RIV EQN **4-1**
Describes how to calculate Lake Ontario outflows (required if routing through Lake Ontario).

WELLAND CANAL DATA **5-45**
Welland Canal discharge from L. Erie.

Appendix B

GLOSSARY

Users are urged to contact the primary author with any suggested additions or changes to this introductory glossary.

Algorithm	A step-by-step procedure for solving a problem or computing a result that accomplishes an objective, then terminates.
Alphanumeric	Consisting of characters (i.e., letters, numbers, and special symbols).
Application	A piece of software used to perform a particular task, such as a word processor or graphics package. Usually, they are geared for a particular segment of users (e.g., business, science, mathematical, and educational applications).
Area/ Elevation Relation	The relation for a given water body used to determine the surface area of the water body given a particular water level elevation.
Argument	A data item supplied to a Fortran statement, function, procedure, or program. They may be constant or variable, and are often used to pass information from one part of a program to another. In Fortran, they may pass to subprograms, from subprograms, or both ways. A synonym for parameter.
Array	A sequentially numbered block of variables with the same type, having the same name, with the individual elements distinguished from one another by subscripts. Fortran arrays may have up to seven subscripts. An array with one subscript is known as a vector, or list. One with two subscripts is called a matrix, or table. Those with three or more are “hypertables”.
ASCII	American Standard Code for Information Interchange. A set of 256 7-bit character codes that most computers use to represent letters, digits, special characters, and other symbols. Only the first 128 codes are standardized. The remaining 128 are special characters defined by the computer manufacturer.
ASCII Text File	A file that can be intelligibly displayed on the monitor or printed.
Assignment	A value assigned to a variable.
Backup	A copy of a file, directory, or disk made for safe keeping, in case the original is lost, deleted, or damaged.
Batch File	A file containing a sequence of commands. When the name of the file is typed at the command line, each command is performed in turn without delay, just as if the commands had been typed individually. The filename is a “command” in itself. Batch filenames must always have the extension “.BAT”.

Beginning-Of-Period	Relating to a value determined to represent the initial value at the beginning of a period (e.g., beginning-of-month values are assumed to be the average of the daily mean values from the last day of the previous month and the first day of the present month).
Binary	Mathematical term referring to the base-two numbering system, which uses only two digits: 0 and 1.
Binational	Of, relating to, or constituting a group or association having members in two nations(for CGLRRM matters, Canada and the United States).
BIOS	Basic Input/Output System. Machine language programs built into a computer which provide the lowest level of operations in a computer system.
Bit	The two-valued datum symbolized by an "on-or-off" memory circuit. An abbreviation for binary digit.
Block	A sequence of declarations, definitions, or statements.
Boot	To start a computer system by loading the operating system from mass storage or another peripheral. An abbreviation of bootstrap from the phrase "pulling oneself up by one's own bootstraps."
Bug	An error in a program or device.
Call	The invocation of a Fortran subroutine from the main program.
CGIP	An acronym for the Chippewa-Grass Island Pool, located just upstream of Niagara Falls on the Niagara River. The stage of this pool is often used as the downstream level in two-gage outflow relationships for the river. An International Joint Commission directive sets this level at 171.16 m.
CGLRRM	An acronym for the Coordinated Great Lakes Regulation and Routing Model. Given the parameters describing the hydraulic characteristics of the system, and hydrologic input data, the model will simulate Lakes Superior and Ontario regulation, and route flows through each of the Great Lakes.
Chicago Diversion	A diversion of water out of Lake Michigan through the Chicago Sanitary and Ship Canal to the Des Plaines River and ultimately to the Mississippi River watershed.
Clipboard	A storage area that holds text being copied or moved.
Code	One or more special modules linked together to form a complete set of instructions that a computer can execute to perform a given task.

Coding	The stage of programming in which an algorithm is expressed in the vocabulary and grammar of a particular programming language.
Command	A short program designed for a specific task.
Command Line	The command followed by any other relevant information given at one time to the computer, to enable it to carry out the instructions. Command lines must be terminated by a carriage return (pressing <Enter>).
Comment Line	A line in a Fortran program marked by "C" or "*" (or "#" in CGLRRM input files) in the first column and, as a result, ignored by the compiler (or program). Such lines are used to provide explanations of the program for human readers. Also called internal documentation.
Compensating Works	The hydraulic control dam straddling the international boundary at Sault Ste. Marie, used in the (partial) regulation of Lake Superior. Completed in 1921, the dam consists of 16 control gates, which span much of the outlet of Lake Superior. Flow through the structure is specified using Plan 1977-A.
Compile	To translate statements in a higher-level programming language (e.g., Fortran) into a lower-level form of instructions that the computer can execute. A compiler performs this process.
Compiler	A program that automatically translates statements in a higher-level language (e.g., Fortran) in a program into a (machine) language understood by computers.
Computer	A programmable machine that manipulates finite symbols, performing logical operations on them. It consists of main memory and control units, including the central processing unit (CPU), but does not include peripherals. Contrast with computer system.
Computer System	A complete system for computing, consisting of hardware components together with the software.
Constant	A value that does not change during program execution.
Consumptive Use	A use of water (e.g., domestic water supply, industrial uses, agriculture, etc.) that does not directly return water to the water body drawn from.
Control Structure	*Hydraulics:* A dam or weir placed at the outlet of a lake or reservoir to permit regulation of outflows and levels. *Programming:* A group of instructions that control the execution of statements by means of conditional execution or loops.
Coordinated	Designed to be developed, constructed, and used together by several groups of people to attain mutual utility and effectiveness, often via the harmonious functioning of several parts.

Dam	A barrier, acting as a hydraulic control structure, built across a watercourse as a means of impounding water. Oftentimes used in hydroelectric power production.
Data	The numbers and text (or combination of the two) processed by a computer during a model run.
Database	A stored collection of data, usually on a mass-storage medium, in a form that allows systematic retrieval and processing (e.g., a data file). Usually the term database is reserved for very large, highly organized files designed for retrieval and processing.
Data File	A file that contains the data used or generated by a program.
Datum	Singular of data. A reference point used as a basis for calculating or measuring.
Debugging	Locating and removing the bugs (errors) that are the source of problems discovered during run-time testing of a program or model.
Default	A pre-set value that is assumed until the user specifically changes it. The conditions that exist when the user does not define values.
Deterministic Forecast	Estimate of the value(s) of variable(s) in the future, performed by simple means (e.g., using historical averages), or sophisticated methods such as modeling of pertinent physical processes applied to current conditions.
Detroit River	The connecting channel between Lake St. Clair and Lake Erie. Approximately 50 km in length.
Deviation	Any departure from the anticipated outflow determined using a regulation plan. Deviations may be unintentional, but are most often as a result of regulatory discretion, emergencies, or winter requirements.
Device	A piece of computer equipment (hardware), such as a monitor or a printer, that performs a specific task. Usually, a device is a peripheral.
Dialog Box	A box that appears when a menu command is selected that requires additional information.
Dimension	The number of subscripts required to specify a single array element.
Directory	The index of files maintained on a disk.
Disk Operating System (DOS)	A collection of programs that manages computer resources and other programs on a computer.

Diversion	A transfer of water from one watershed into another.
Double Precision	A real (floating-point) value that occupies eight bytes of memory and are accurate to 15 or 16 digits.
Drainage Basin	A geographical area or region bounded by a water divide and draining ultimately to a particular watercourse or body of water.
End-Of-Period	Relating to a value determined to represent the initial value at the end of a period (e.g., end-of-month values are assumed to be the average of the daily mean values from the last day of the present month and the first day of the next month).
Error	An act or statement involving an unintentional, ignorant, or imprudent deviation from the truth or accuracy, that fails to achieve the desired effect. Something produced when a mistake is made.
Error Message	A usually fatal advisory message issued by a compiler or program. More serious than a warning message.
Evaporation	Loss of water from a surface by the expulsion of liquid water that has been converted into a vapor.
Exceedance Probability	The chance that a given value will exceed values of the past (e.g., There may be a 95% chance that the water level of Lake Superior will exceed 183.0 m. The exceedance probability of this event is 95%).
Executable	A file with the extension .EXE, .COM, or .BAT that can be run by a computer by typing the file name at the command line.
Execute	To run a program or model or carry out a specific statement at the command line.
Expression	A combination of operands and operators that yields a single value.
Extension	The part of a filename following the period (e.g., .OUT is the extension in the file name TEST.OUT).
FAQ	An acronym of the phrase, "frequently asked questions."
Field	A portion of a line of input or output.
File	A collection of related instructions or data stored on disk.
Filename	The name assigned to a file. A period (.) separates the filename from the extension.

Filespec	Short for file specification, filespec is the complete combination of drive letter, path name, filename, and extension that uniquely identifies a file.
Floating Point	Notation, similar to scientific notation, that is a convenient way of expressing very large or small numbers or fractions. They have two parts: an exponent and a fractional part (the mantissa).
Forecast	To predict future conditions (e.g., outflows and levels) by rational study and analysis of available pertinent data or correlated hydrometeorological observations (i.e., deterministic forecasting). Oftentimes, used to estimate the likelihood of occurrence of certain events or conditions in the future (i.e., probabilistic forecasting).
Fortran	An algebraic, high-level programming language designed in the mid-1950's by a team headed by IBM's John Backus, and intended for numeric calculations such as those in science and engineering. Originally written FORTRAN, an acronym for "Formula Translation".
Fortran 77	A standardized version of Fortran, ANSI X3.9-1978, as defined by the American National Standards Institute in 1977-78.
Function	A subprogram that returns a simple value associated with its own name.
Gated	Related to the Lake Superior outflows that pass through the gates of the binational Compensating Works hydraulic control structure located at Sault Ste. Marie.
GLERL	An acronym of the Great Lakes Environmental Research Laboratory, Ann Arbor, Michigan.
Great Lakes	A chain of five large lakes (Superior, Huron, Michigan, Erie, and Ontario) located in central North America that border one Canadian province and eight U.S. states and drain through the St. Lawrence River to the Atlantic Ocean. The smaller Lake St. Clair (located downstream of Lake Huron and upstream of Lake Erie) is often considered as well. Lakes Michigan and Huron (as well as Georgian Bay) are often considered to act together hydraulically.
Groundwater	Water lying in underground storage in the portion of the earth that is wholly saturated.
GUI	An acronym for the phrase, "graphical user interface." A series of one or more on-screen windows with which a user can interact. Normally used as a means of input and/or output. Most GUIs are designed using standards developed for all Windows applications.
Hardcoded	A program unit which is coded so that it cannot readily be changed by users without modification and recompilation of the program code itself.

High-Level Language	A programming language that is relatively similar to normal human communication.
Hydraulics	The earth science that deals with the practical applications of liquid (e.g., water) in motion. Of or relating to hydraulics.
Hydrology	The earth science that deals with the waters of the earth, their occurrence, circulation, distribution, chemical and physical properties, and their relation to living things and the environment.
Ice	A decrease in the flow in a channel due to the presence of ice. Generally, such outflow restrictions result in a temporary increase in upstream water levels.
IJC	An acronym for the International Joint Commission. A binational body established within the 1909 *Boundary Waters Treaty*. Of the six members, the U.S. President appoints three Americans (with the advice and approval of the Senate), and the Governor in Council of Canada appoints three Canadians, on the advice of the Prime Minister. Commissioners must follow the Treaty as they assist governments in trying to prevent or resolve disputes related to water bodies that lie along, or flow across, the border. The Commission has set up many expert boards to help it carry out its responsibilities.
Increment	The number of intermediate calculations within each routing period within a program or model. The timestep is the routing period divided by the increment. In general, the larger the number, the more precise the results. See also *timestep* and *routing period*.
Information	The human significance of data. The reception of knowledge or intelligence.
Input	The data that a program reads. Usually input is either read from disk by a program or typed in at a keyboard.
Input File	A file from which a program reads (input) data.
Input/Output	A term that refers to the devices and processes involved in the computer's reading (input) and writing (output) data.
Integer	A whole number. Integers can have a range from –32,768 to +32,767.
Interface	The boundary between two systems or entities, such as a monitor and the computer, or a user and a program.
Internal Documentation	See comment line.

International	Of, relating to, or constituting a group or association having members in two or more nations.
Iterative Method	A process whereby actions are repeated. When used in the CGLRRM, the middle lakes simultaneous (routing) equations are solved via a repetitive process until tolerance values have been reached.
Kilobyte (KB)	A unit of memory in a computer denoting 1024 bytes. 1024 KB=1 MB (Megabyte).
Logic Error	A program or model error in which the computer is instructed to do something inappropriate from the user's point of view. A type of semantic error.
Long Lac Diversion	A diversion of water from the Kenogami River watershed (within the James Bay watershed), into Long Lake, which drains into Lake Superior.
Loop	A control structure that repeatedly executes the same block of statements until a condition or conditions are satisfied.
Machine Language	Instructions that a microprocessor can execute. Consists solely of bit patterns.
Main Program	In a program that calls subroutines or subprograms, the calling program is the main program. The first program to run.
Maintenance	Work that continues on a program or model after it is written and first run, to maintain or enhance its usefulness. This may involve adding features, revising current ones, debugging, and transportation to new computer systems. Maintenance comprises the bulk of the costs in model operation.
Megabyte (MB)	A unit of memory in a computer denoting 1024 Kilobytes. 1024 MB=1 GB (Gigabyte).
Memory	Where computers store data and programs. Measured in bytes (8 bits), since a byte of memory is enough to represent a character. A bit is the smallest unit of memory, and has the value 0 or 1.
MIDLAKES	A program developed at GLERL to solve the nine simultaneous equations for routing through Lakes Michigan, Huron, St. Clair, and Erie using an algorithm based on matrix algebra.
Model	A computer program serving as a representation or simulation, presented as a system of data and inferences to describe a physical entity.
Module	A discrete unit of code in a program.

Nested Loops	A control structure in which at least one loop is contained within another.
Net Basin Supplies	Abbreviated NBS. The net balance of water supplied to a drainage basin but not including inflows from diversions or connecting channels draining into the inlet of the lake. In general, considered the sum of basin gains (i.e., over-lake precipitation and basin runoff (including groundwater)) minus basin losses (i.e., evaporation).
Net Total Supplies	Abbreviated NTS. The net balance of water supplied to a drainage basin, including inflows from diversions and connecting channels draining into the inlet of the lake. In general, considered the sum of connecting channel inflows and net basin supply.
Niagara River	The connecting channel between Lake Erie and Lake Ontario. Approximately 58 km in length. Includes Niagara Falls, a waterfall with a drop of approximately 50 m. Total fall is about 100 m from Lake Erie to Lake Ontario.
Null Character	The ASCII character encoded as the value 0.
Ogoki Diversion	A diversion of water from the Ogoki River watershed (within the James Bay watershed), out of the Ogoki Reservoir into Lake Nipigon, which drains into Lake Superior.
Online	A computer or peripheral is online when it is switched on and ready to accept information.
On-line	When a person is browsing (or “surfing”) the Internet (or “Web” (short for World Wide Web)) at a computer, he/she is on-line.
Operand	A constant or variable value that is manipulated in an expression.
Operator	One or more symbols that specify how the operand(s) of an expression are manipulated.
Ottawa River	A river forming much of the southern border between Ontario and Quebec that flows for approximately 1120 km between the provinces before emptying into the St. Lawrence River just upstream of Montreal.
Outflow	Also often referred to as flow or discharge. The amount of water passing through a watercourse, generally taken as the volume of water passing some point in a given period.
Output	The data that results from processing a program or model, based on some form of input data.

Overflow	An error that occurs when a value assigned to a numeric variable falls outside the allowable range for that variable's type.
Parameter	An independent variable symbol in a statement that is replaced by actual variables and values when a function or subprogram is invoked. Its value determines some characteristic of a system. A synonym for argument.
Parity	A means of checking that data was transferred correctly, either between computers across a communications link, or between components inside a single computer, such as a disk and memory.
Path Name	The complete specification that gives the location of a file. They usually include a drive letter followed by a colon, one or more directory names, and a file name.
PC	An abbreviation for "personal computer".
Peripheral	An external device connected to a computer, generally used for input and output or storage (e.g., disks, modems, printers).
Plan 1958-D	The currently adopted regulation plan for Lake Ontario regulation, in use since 1963. Artificial control of the outflows and levels of Lake Ontario must follow this plan in an attempt to satisfy the criteria and other requirements (established by the International Joint Commission in 1956) to protect or to provide advantages to various interests concerned.
Plan 1977-A	The currently adopted regulation plan for Lake Superior regulation, in use since 1990. Artificial control of the outflows and levels of Lake Superior must follow this plan in an attempt to give consideration to downstream conditions while satisfying the criteria (established by the International Joint Commission in 1914, as amended) to protect or to provide advantages to various interests concerned. The fundamental principle is to keep the levels of Lakes Superior and Michigan-Huron at the same relative position with respect to each other's long-term monthly mean levels.
Portability	The ability for software programs to be able to run successfully and easily on most current computer systems without the need for significant modifications beforehand.
Precipitation	A deposit of hail, mist, rain, sleet, or snow on the surface of the Earth (either land or water). Also, the quantity of water so deposited.
Present Working Directory	The directory in which the model first looks for files. File "CGLRRM.INI" for the current run must preside here.
Probabilistic	An estimate of the value(s) of variable(s), with each value or set of values

Forecast	associated with an estimated probability of some kind, reflecting the anticipated likelihood of an event defined in terms of the value(s) (e.g., an exceedance event).
Program	Software. One or more specially coded modules linked together to form a complete, ordered list of instructions that a computer can execute to perform a given task. Each program represents an algorithm.
Prompt	An output request displayed by the computer for the user to provide information or perform an action (input).
Quarter-Monthly	Occurring four times per month on prescribed days. In February, the four quarter months are evenly spaced (seven days each (1-7, 8-14, 15-21, 22-28); forth quarter has eight days in leap years, though (22-29)). For all other months, the first quarter is eight days (1-8), the second quarter is seven (9-15), the third quarter is eight days (16-23), and the forth quarter is seven or eight days (24-30 or 31).
Real	A floating-point number with a fractional part, stored in a kind of scientific notation, with sign, fractional part, and exponent.
Record	A group or line of related data, not necessarily of the same type, written or read as a group.
Retardation	A decrease in the mean velocity in a channel due to an impediment to flow (e.g., ice, weeds, or obstructions). Generally, such outflow restrictions result in a temporary increase in local water levels.
Regulation	Artificial control of a lake's outflow and levels, usually as per a plan adopted to protect or provide advantages to the various interests concerned. Normally, dams are employed as the control structure used to affect regulation at the downstream end of the regulated lake.
Roughness	Generally taken as the Manning's equation roughness factor (a measure of the degree of impedance to flow caused by a submerged surface) for ice cover along a river reach (over and above the value assumed for the river bed). The more jagged the underside of the ice cover becomes, the higher the roughness associated with it.
Rounding	Approximating a numeric value so that its storage representation in a finite number of digits will be as accurate as possible.
Roundoff Error	The error caused when a number is rounded for storage.
Routine	Executable code residing within a module that usually represents a particular feature or process.

Routing	To send water by selected routes through a waterway's lakes and rivers.
Routing Period	The period of time between successive determinations within a program or model. Used to determine the frequency of output. See also *increment* and *timestep*.
Runoff	The portion of precipitation on land that ultimately reaches a body of water (especially that which flows overland to rivers, or that which enters a reservoir via rivers (then referred to as stream runoff)).
Run Time	The time during which a program is executing. Run time refers to the execution time of a program, rather than to the execution time of the compiler.
Run Time Errors	Errors that can only be detected while the program is running.
St. Clair River	The connecting channel between Lake Huron and Lake St. Clair. Approximately 64 km in length.
St. Lawrence River	The connecting channel between Lake Ontario and the Atlantic Ocean. Approximately 1223 km in length.
St. Mary's River	The connecting channel between Lake Superior and Lake Huron. Approximately 101 km in length.
Semantic Error	An error in the meaning, as opposed to syntax. Semantic errors cause programs to run incorrectly.
Sidechannel	Relating to the outflows from Lake Superior over and above those not passed through the Compensating Works. Included are outflows for power production and lockages, as well as domestic and industrial uses.
Simulation	The imitative computer representation of the functioning of a system or process by means of the functioning of another. Oftentimes, used as a means of estimating historic conditions by modeling known physical data within limited periods of time in the past.
Source Code	Code written in a higher-level language (e.g., Fortran) to be submitted to a compiler for translation to a lower level (machine code). The result of the translation is called object code.
Stage-Discharge Relation	The unique relation between stage (level above a given datum) and discharge (outflow) at a cross-section of a channel reach defined by a "rating" curve. In general, they take the form: $Q = a\,(h - h_0)^{b}$

Where Q = discharge or outflow,
h = stage,
h_0 = datum elevation, and
a, b are constants.

Stage-Fall-Flow Relation The unique relation between stage (level about a given datum), fall (difference in level between locations) and discharge (outflow) over a length of channel reach between two locations. In general, they take the form:

$$Q=a(h_1-h_0)^b(h_1-h_2)^c$$

Where Q = discharge or outflow,
h_0 = datum elevation at location 1,
h_1 = stage at location 1 (upstream location),
h_2 = stage at location 2 (downstream location), and
a, b, and c are constants.

Statement A combination of one or more Fortran keywords, variables, and operators that the compiler treats as a single unit. A statement can stand by itself on a single program line.

String A sequence of characters stored together (often identified by a symbolic name) which may be a constant or a variable.

Subprogram A program unit that cannot execute by itself. Functions or subroutines that are not part of the main program.

Subroutine A subprogram invoked with a call from the main program that returns values by means of an argument list.

Subscript An index used to identify elements of an array.

Syntax Error An error making a statement grammatically incorrect and therefore, untranslatable into machine code. Compilers report such errors, and the CGLRRM's error checking routine usually finds these.

Text Ordinary, printable characters including the uppercase and lowercase letters of the alphabet, the numbers 0 through 9, and punctuation marks.

Text Editor A simple program that is used to create, edit, save, and view ASCII text files on a computer. Similar to word processors, editors usually do not have the ability to justify, control page breaks, etc.

Text File A file containing only ASCII text.

Timestep	Denoted dt. The period of time between successive computations within a program or model. In general, the smaller the timestep, the more precise the results are. The timestep is the routing period divided by the increment. See also *increment* and *routing period*.
Tributary	A water course which drains into another body of water.
Type	The numeric format of a variable, which may be user-defined.
USACE	A common acronym of the United States Army Corps of Engineers.
Variable	A value that can, and usually does, change during program execution.
Verbosity	A colloquial term related to wordiness or long-windedness. A family of CGLRRM parameters used to control the level of output in the detailed log for each run of the model.
Warning Message	A non-fatal advisory message issued by a compiler or program. Not as serious as an error message.
Water Level	The elevation of the water surface of a given body of water above a certain datum.
Weed	A decrease in the flow in a channel due to an overabundance in weed growth along the riverbed. Generally, such outflow restrictions result in a temporary increase in upstream water levels.
Weir	A partial dam, acting as a hydraulic control structure, in a watercourse often used to raise upstream water levels or to divert flow. Under normal operating conditions, water is always passing over a weir.
Welland Canal	A diversionary shipping canal from Lake Erie to Lake Ontario constructed as part of the St. Lawrence Seaway project to permit shipping traffic to bypass Niagara Falls through a series of locks. Also supplies water for hydropower generation.
Window	An area on the screen used to display part of a file or to enter statements. It refers only to the area on the screen, and not to what is displayed in the area.

Appendix C

ERROR MESSAGES

The following is a complete record of the error and warning message database for the CGLRRM (contents of default filename “MESSBASE.TXT”). The program uses the following text blocks to convey a message to the user whenever a suspect piece of parsed data is encountered.

1000 Tried to open a file (for write) and was denied. There may be a permission problem with writing to the present working directory or the specific filename. Another process may already have the file open. The filename, IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), and the build information are as follows:
1001 Could not find the master input file in the present working directory. It contains the names of parameter files to open and read.
1002 The master input file exists, but could not be opened. Another process (e.g., an editor) may already have it opened for write access. The filename, IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), and the build information are as follows:
1003 Experienced a read error. The filename and resulting IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), and the build information are as follows:
1004 A problem was found in the master input file. This file should contain only comments and parameter file names. The program tried to find a file with the name below and failed. Check the spelling and path from the present working directory.
1005 The parameter file exists, but could not be opened. Another process (e.g., an editor) may already have it open for write access. The filename, IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), and the build information are as follows:
1006 A parameter file appears to contain more than 32766 lines. This file may be corrupt. The file in question is named:
1007 Could not find a ':' in one of the lines of a parameter file. A colon is used to separate the parameter ID code from the comma-delimited list of data for that parameter. The line in question reads:
1008 Found an ':' inappropriately located in one of the lines of a parameter file. A colon is used to separate the parameter ID code from the comma-delimited list of data for that parameter. In this case, the parameter ID is shorter than six characters (the shortest one defined). The line in question reads:
1009 Didn't find anything after the ':' on a parameter line. A colon is used to separate the parameter ID code from the comma-delimited list of data for that parameter. In this case, there was no data on the line. The line in question reads:
1010 Did not recognize a parameter ID code. It must be one of those described in the documentation. Check the spelling. The line in question reads:
1011 Found a match error in the list of parameters. Since some parameter codes are concatenated into strings, the end of one code may be combined with the beginning of another code, creating a valid match. Change or re-arrange the codes, and re-compile. The line in question reads:
1012 The parameter matched one in the list, but there was no corresponding label in the list for the computed GOTO. Check any relevant changes made to the source code. The parameter ID code, its index number, and the parameter line it was
read from are as follows:
1013 Could not find the message database file specified by parameter MESSAGEDATABASE. The filename used and the parameter line it was read from are as follows:
1014 Invalid value specified for a VERBOSITY parameter. The character after the ':' must be an integer between '1' and '9'.
1015 Unable to open a temporary file in the output directory specified by the parameter OUTPUTDIRECTORY. The directory may not exist, there may be a permission problem, or the path from the present working directory may be incorrect. The filename, IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line containing the OUTPUTDIRECTORY parameter are as follows:
1016 Was able to open a temporary file in the output directory specified by the parameter OUTPUTDIRECTORY, but could not write to it. There may be a permission problem, or the file may be in use by another process. The filename, IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the

CGLRRM executable), the build information, and the line containing the OUTPUTDIRECTORY parameter are as follows:
1017 Was able to open a temporary file in the output directory specified by the parameter OUTPUTDIRECTORY, and write to it, but couldn't delete it afterwards. There may be a permission problem, or the file may be in use by another process. Be aware that some file systems (such as that used on Novell Netware servers) have separate permissions for write and delete. The filename, IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line containing the OUTPUTDIRECTORY parameter are as follows:
1018 Filename extensions cannot exceed six characters (including the optional leading '.').
1019 Did not recognize the directory separator used as the final character in the string specified for the parameter OUTPUTDIRECTORY. The program successfully opened and wrote to a file, but the directory used may not be correct. Check file time and data stamps in the output directory to ensure that they are not from previous runs. If they are old, check in
the present working directory (or the directories within the path specified in OUTPUTDIRECTORY) for misnamed files.

If the last character in OUTPUTDIRECTORY intentionally isn't a directory separator, then an invalid filespec may be created, and future runs may result in errors.

Otherwise, the model has been compiled on some operating system (or with a file system) with directory separators unknown to the program. Update the DIRSEP array in BLOKDATA.FOR. In this case, please also contact the authors, so that they can update the distribution code.
1020 Could not interpret year given for parameter "Start Date". The first characters after the ':' must be 1-6 digit integers with digits between '0' and '9'. The IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line in question, are as follows:
1021 The specified year is out of range. Must be integer value between 1 and 32766. The value read and the input line are as follows:
1022 Warning. Found no data in the 'month' field of the "Start Date" record. January 1 was assumed as the starting month and day.
1023 Could not interpret month given for parameter "Start Date". The first characters after the first ',' must be 1-2 digit integers between '1' and '12'. The IOSTAT (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line in question, are as follows:
1024 The specified month value is out of range. Must be an integer between '1' and '12'. The value read and the input line are as follows:
1025 Warning. Found no data in the 'day' field of the "Start Date" record. The first was assumed as the starting day.
1026 The specified day is out of range. Must be an integer of at least 1 and less than or equal to the length of the month. The value read and the input line are as follows:
1027 No starting year was given in "Start Date" parameter. This is required information. The input line is as follows:
1028 No ending year was given in "End Date" parameter. This is required information. The input line is as follows:
1029 Could not interpret year given for date parameter. The first characters after the ':' must be 1-6 digit integers with digits between '0' and '9'. The IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line in question, are as follows:
1030 Warning. Found no data in the 'month' field of the "End Date" record. December 31 was assumed as the ending month and day.
1031 Warning. Found no data in the 'day' field of the "End Date" record. The final day of the month was assumed to be the ending day.
1032 . . . assuming starting month and day of January 1
1033 . . . assuming starting day of 1st day of month
1034 . . . assuming ending month and day of December 31
1035 . . . assuming ending day as last day of month
1036 Could not interpret month given for date parameter. The first characters after the first ',' must be 1-2 digit integers between '1' and '12'. The IOSTAT (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line in question, are as follows:

1037 Could not interpret day given for parameter "Start Date". The first characters after the first ',' must be 1-2 digit integers between '1' and the number of days in the month. The IOSTAT (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line in question:
1038 Could not interpret day given for date parameter. The first characters after the first ',' must be 1-2 digit integers between '1' and the number of days in the month. The IOSTAT (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line in question, are as follows:
1039 An Area-Elev record specifies no area (blank first field). This is required information. The input line is as follows:
1040 Unable to read a float from the following string because it is longer than 32 characters long. The input line is as follows:
1041 Failed to recognize the units (if any) specified for a parameter value. Please consult the documentation for a comprehensive list of allowable units. The field in question and the input line are as follows:
1042 Recognized the units specified for a parameter value, but did not find a number in front of the unit label. The field in question and input line are as follows:
1043 Could not read a parameter value in as a REAL number. Check the entry (may have O's instead of 0's, etc.). Some compilers have specific requirements (e.g., M$ FORTRAN 5.0 compilers do not accept exponential formats without an exponent (e.g., 1.234E), SparcStation compilers assume zero (i.e., 1.234E0). The IOSTAT (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line in question, are as follows:
1044 Inappropriate input units used, resulting in an unrecognized or illegal unit conversion (e.g., a lake level specifying "176.54 m3s" in lieu of "176.54 m)". Program terminated. Note that all computations are performed in metric units. English units may be specified, but are immediately converted to metric. Metric-to-metric conversions are also performed (e.g., square kilometers to square meters). The units specified, the units the program intends to work with, and the input field, are as follows:
1045 Too many surface area/elevation data pairs specified. The default maximum is five points to define the curve. If more points are desired, change the second dimension on array AVSZ, adjust the limit in function READAVSZ, and re-compile.
1046 Encountered ambiguity reading a parameter value. The last five characters matched the tens of cubic meters per second label ("10m3s"), but there is no separation between the number and units (e.g., 210m3s can be interpreted as either 20 or 210 m3s. The line in question is as follows:
1047 Encountered inconsistency with area-elevation data. Read the first point of a valid data pair, but a constant area had already been specified. The line in question is as follows:
1048 Encountered inconsistency with area-elevation data. Read a constant area, but a first area/elevation point had already been read. The line in question is as follows:
1049 Encountered mistake in area-elevation data. For two of the points, area is decreasing with elevation. The data in question are as follows:
1050 Encountered mistake in area-elevation data. Two of the points have the same elevation. Points must be at least a millimeter apart in elevation. The data in question are as follows:
1051 Tried to interpolate/extrapolate a value from two points, but the X values for each point were practically identical. This would result in a division by zero error. The X values data in question are as follows:
1052 Warning. Extrapolated based on the closest pair of data points. It is recommended that the curve be extended. The extrapolated value and data points used are as follows:
1053 Discharge equation record specifies no 'K' value (blank first field). This is required information, corresponding to the outflow equation coefficient, in the general discharge equation:

Q= K * (WT*ZU + (1.-WT)*ZD - YM)**A * (ZU-ZD+ZC)**B

The input line is as follows:
1054 Discharge equation record specifies no 'YM' value (blank second field). This is required information, corresponding to the outlet bottom elevation constant, in the general discharge equation:

Q= K * (WT*ZU + (1.-WT)*ZD - YM)**A * (ZU-ZD+ZC)**B

The input line is as follows:

1055 Discharge equation record specifies no 'A' value (blank third field). This is required information, corresponding to the exponent for the "depth" term, in the general discharge equation:

Q= K * (WT*ZU + (1.-WT)*ZD - YM)**A * (ZU-ZD+ZC)**B

The input line is as follows:
1056 Discharge equation record specifies no 'B' value (blank fourth field). This is required information, corresponding to the exponent for the "fall" term, in the general discharge equation:

Q= K * (WT*ZU + (1.-WT)*ZD - YM)**A * (ZU-ZD+ZC)**B

The input line is as follows:
1057 Discharge equation record specifies no 'WT' value (blank fifth field). This is required information, corresponding to the weighting factor for the upstream lake, in the general discharge equation:

Q= K * (WT*ZU + (1.-WT)*ZD - YM)**A * (ZU-ZD+ZC)**B

The input line is as follows:
1058 Discharge equation record specifies no 'ZC' value (blank sixth field). This is required information, corresponding to the fall adjustment constant, in the general discharge equation:

Q= K * (WT*ZU + (1.-WT)*ZD - YM)**A * (ZU-ZD+ZC)**B

The input line is as follows:
1059 Parameter record neglected to specify a value (blank first field). Please edit the record. The input line is as follows:
1060 An MIDLAKEINC, SUPERIORINC, ONTARIOINC, or P77AMIDLAKEINC record specifies an invalid value. The number of increments must be between 1 and 999. The value read and the input line are as follows:
1061 Encountered read error. The IOSTAT value which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line in question, are as follows:
1062 Lake Superior level fell too low to use the gate equations. It is likely that the level has been outside the range of validity for some time, and the discharge values should be viewed with skepticism. Check model input. The level is as follows:
1063 Failed to specify a necessary starting elevation.
1064 Could not find the data file specified. The filename used and the parameter line it was read from are as follows:
1065 Tried to compute a discharge equation using water levels outside the range for which it is valid. Downstream water level is higher than upstream water level (plus ZC), or the upstream water level is lower than the channel invert (second value on the parameter input line describing the equation).

This may happen when routing through the middle lakes with really large and abrupt changes in the supply or retardation values, because the level of Lake St. Clair will fluctuate too much. Check input data, then increase the number of increments per period. The levels in question are as follows:
1066 Parameter ID code is longer than 16 alphanumeric characters. Check your parameter line against the recognized codes in the documentation. The line in question is as follows:
1067 Found a ':', but no parameter ID code before it. Check input file. The line in question is as follows:
1068 Tried to open a file (for write) and was denied. The file should hold summarized input data. There may be a permission problem with writing to the present working directory or the specific filename, or another process may already have the file open (e.g., another instance of the model currently running). The filename, IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), and the build information are as follows:
1069 Found keyword 'INTERVAL' in a time-series data file, but could not determine the value after it. The next letter after the keyword must be 'D', 'W', 'Q', or 'M' (daily, weekly, quarter-monthly, or monthly). The line in question is as follows:
1070 First word begins with a 'U' or 'I', but it does not match either of the keywords required in time-series data files (i.e., INTERVAL' or 'UNITS'). Check the entry. The line in question is as follows:

1071 Found a line that appears to be a comment, but it did not start with a '#'. Check the entry. The line in question is as follows:
1072 No units specified in time-series data file. File must contain a line such as:
Units: m3s
1073 No data found in a time-series data file. Program terminated.
1074 No time interval specified in time-series data file. File must contain a line such as:
Interval: quarter-monthly
1075 Found data, and read the units and interval from time-series data file, but could not backspace it to the first data line.
1076 Failed to parse a line of quarter-monthly input data according to the standard format (I4,I2,12F8). File may be in some other format, was corrupted, or has missing lines. Regardless of the starting quarter-month and ending quarter-month, all four lines for each year in the simulation must be present.
1077 Failed to parse a line of monthly input data according to the standard format (I4,12F8). Check the year value and look for corrupted data. The IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line in question, are as follows:
1078 Failed to parse a line of monthly input data according to the standard format (I4,12F8). The year given on the line was not sequential from that of the previous line. File may be in different format, have missing lines, or was corrupted.
1079 Reached end of file (EOF) before all necessary data read. Several possible reasons: incorrect ending date, data must be appended to the file, or the wrong file was specified. Note:

a) Implementation of when the EOF is encountered during file access varies among computer systems (e.g., on SPARCstations, if the last line of a data file lacks a linefeed, EOF is reached, even though the line has valid data). If no mistake is found, append a carriage-return (for DOS compatibility) or add a blank line to the end of data files.

b) If a weekly routing interval is used, the end of the final period is actually the end day of the week containing the ending date specified in the run parameters. The beginning of a week is taken as the day of the starting date (e.g., a weekly run specified to start on Monday, 5Jan1998, and end on 31Mar1998, will actually require data through Sunday, 5Apr1998. Each period is Monday to Sunday, and the final period is the one containing the ending date. If data for Apr1998 was unavailable, the ending date should be changed to Sunday, 30Mar1998).
1080 Time-series input data file specified inappropriate units. Must be m3s, cfs, tcfs, or 10m3s.
1081 Warning. Solving for level of Lakes Michigan-Huron without specifying net basin supplies. 0's will be assumed.
1082 Warning. Solving for level of Lake St. Clair without specifying net basin supplies. 0's will be assumed.
1083 Warning. Solving for level of Lake Erie without specifying net basin supplies. 0's will be assumed.
1084 Warning. Solving for level of Lakes Michigan-Huron without specifying Chicago diversion data. 0's will be assumed.
1085 Warning. Solving for level of Lake Erie without specifying Welland Canal flow data. 0's will be assumed.
1086 Error reading summary inflow file.
1087 Error converting a line read from summary inflow file to floating point numbers.
1088 Routing interval specification record specifies an invalid value. The first character after the ':' must be a 'D', 'W', 'Q', or 'M' (daily, weekly, quarter-monthly, or monthly routing through the middle lakes). The value read and the input line are as follows:
1089 Tried to open a file (for read) and was denied. The file should hold summarized input data. There may be a permission problem writing to the present working directory or the specific filename. Another process may already have the file open (e.g., another instance of the model currently running). The filename, IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), and the build information, are as follows:
1090 The simultaneous solution of routing equations for a given increment fell below an arbitrary accuracy threshold at least once. The continuity test for a period is based on arithmetically summing the water balance terms (outflow, inflow, diversions, net basin supply, and average rate of change in storage). The message is triggered when this result exceeds a threshold value, determined according to FlowRound. If FlowRound is 1, then continuity must be satisfied to within 10 m3s. If FlowRound is 0, then continuity must be satisfied to within 1 m3s. This often results from rounding effects on lake levels. Occasionally it occurs due to failings in the mathematics. In this case, it can usually be fixed by using more increments. Consult file detailed.log to get more information regarding each occurrence.

1091 Could not converge during iterative routing procedure. The simultaneous equations could not be solved to the tolerances specified within the iterations allowed. Review end of file detailed.log. Increase number of routing increments, increase the flow rounding tolerance (default could be too low), or decrease the level rounding tolerance (e.g., rounding to nearest cm may result in lake levels that are 1 cm apart, but yield flows that are farther apart than the tolerance permits).
1092 A MIDLAKESOLUTION record specifies an invalid value. The first character after the ':' must be an 'I' for iterative procedure (as in Plan77A documents), or 'M' for the matrix-based method (i.e., GLERL's MIDLAKES model). The value read and the input line in question are as follows:
1093 If the run starts after the first of the month, the initial levels for Lakes Superior, Michigan-Huron, St. Clair, and Erie can't be used by Plan 1977A as beginning-of-month levels. BOM levels must be specified with the parameters "Sup BOM Level", "MHU BOM Level", "St C BOM Level", and "Eri BOM Level".
1094 A PREVSTMRIVFLOW record specifies no initial Lake Superior outflow for the previous month (blank first field). The input line is as follows:
1095 Specified an ending date earlier than the starting date. The values read are as follows:
1096 Data file does not include data back to the starting date specified. The filename and the date at which data begins are as follows:
1097 Tried to open an output file (for write) and was denied. There may be a permission problem with writing to the output directory or the specific filename. Another process may already have the file open. A name was constructed from the specified (or default) output directory, a 7- or 8-character root name assigned by the program, and from the specified (or default) output file extension. Check the validity of the resulting filename on the file system and operating system in use (e.g., '&' can be used with Win95 but not Windows NT). The filename constructed, the IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), and the build information, are as follows:
1098 MonthLength record specifies an invalid value. The first character following the ':' must be either an 'A' for actual, or a 'U' for uniform.
1099 Specified a non-zero exponent for the downstream level in the Niagara River flow equation, indicating use of a two-gage discharge relationship, but also specified use of the "matrix" solution for solving the simultaneous equations on the middle lakes. The two-gage equation in the matrix solution requires a starting level for the Chippewa-Grass Island Pool. Insert a parameter file line such as:

CGIP Start Level: 171.16 m
1100 Specified a non-zero exponent for the downstream level in the Niagara River flow equation, indicating use of a two-gage discharge relationship, but also need to specify a time-series of levels to use for the Chippewa-Grass Island Pool. These can be either read from a file (specified with parameter CGIPAVELEVELDATA), or assigned from a repeating sequence. Insert a parameter file line such as:

CGIP Ave Level Data: ../tsdata/niamlev.act
1101 Warning. Specified an exponent of zero for the downstream level in the Niagara River flow equation, indicating use of a single-gage discharge relationship, but also specified a starting level for the Chippewa-Grass Island Pool (not required). Check for redundant input parameters.
1102 Warning. Specified an exponent of zero for the downstream level in the Niagara River flow equation, indicating use of a single-gage discharge relationship, but also specified a time-series data file for the Chippewa-Grass Island Pool (not required). Check for redundant input parameters.
1103 Had difficulty locating first comma-delimited field in the first line of a repeated time-series data set.
1104 Failed to recognize the units (if any) specified for some repeating time-series data. Please consult the documentation for a comprehensive list of allowable units. The parameter ID portion of the line in question is unavailable, but the remainder is as follows:
1105 No time interval specified on the first line of a repeating data set. After specifying the units, a field starting with 'd', 'w', 'q', or 'm' is required. The parameter ID portion of the line in question is unavailable, but the remainder is as follows:
1106 Invalid time interval specified on the first line of a repeating data set. After specifying the units, a field starting with 'd', 'w', 'q', or 'm' is required. The parameter ID portion of the line in question is unavailable, but the remainder is as follows:
1107 Could not parse a comma-delimited field from the string listed below. The third field (or first, if a continuation of the data from a prior line) must show the number of values contained on the rest of a line for repeating data. The parameter ID portion of the line in question is unavailable, but the remainder is as follows:

1108 Tried to read an integer from the string listed above. Field must show the number of values contained on the remainder of a line for repeating data. Either could not be read as an integer, or value is < 1. The parameter ID portion of the line in question is unavailable, but the remainder is as follows:
1109 Specifying either too many or too few values for a repeating time-series. The parameter ID portion of the line in question is unavailable, but the remainder is as follows:
1110 Tried to read a floating point from the string listed above. Field must contain one value from a repeating time-series sequence. The parameter ID portion of the line in question is unavailable, but the remainder is as follows:
1111 Could not parse a comma-delimited field from the string listed below. Field is part of a comma-delimited list of repeating time-series values. The parameter ID portion of the line in question is unavailable, but the remainder is as follows:
1112 Specified a non-zero exponent for the downstream level in the Niagara River flow equation, indicating use of a two-gage discharge relationship, but also specified use of the "iterative" solution for solving the simultaneous equations on the middle lakes. The two-gage equation in the iterative solution doesn't need a starting level for the Chippewa-Grass Island Pool. Comment this parameter out.
1113 Warning. Parameters indicate solving for at least one of the middle lakes, but not all of them. Must solve for all three. If any of 'M', 'C', or 'E' appear in the "Lakes to solve for" parameter, they must all be there. Failed to specify solving for the following lakes:
1114 Warning. Specified a starting level for a lake, but didn't identify it as one of the lakes to solve for. Check input consistency.
1115 Warning. Specified NBS data for a lake, but didn't identify it as one of the lakes to solve for. Check input consistency.
1116 Rounding-related record specifies an invalid value. The value specified determines the decimal place to round to, and must be between -6 and 3. If a value of -99 is specified, no rounding is performed, and the full 64-bit floating-point representation is carried forward in the computations. The values read are as follows:
1117 A STMRIVSOLUTION record specifies an invalid value. This parameter tells how to compute St. Mary's River flows (Lake Superior outflows). If '1977A' appears after the ':', then the program uses Plan 1977-A to calculate Superior outflows. If '1955' appears after the ':', then the program uses the 1955 Modified Rule of 49 to calculate Superior outflows. If 'O' appears, an outflow equation is used (that must be specified using the STMRIVEQN parameter; e.g., the pre-project equation). The value read and the input line are as follows:
1118 Note: performing non-monthly Lake Superior routing using Plan 1977-A. Please refer to assumptions in documentation for Superior modeling.
1119 Superior increment record specifies no integer value (blank first field). Edit the record. The input line is as follows:
1120 Lake Superior listed as one of the lakes to solve for, but failed to specify how to compute Lake Superior outflows. Insert a STMRIVSOLUTION record.
1121 Warning. Time step for middle lakes routing is extremely large. Recommend reducing time step.
1122 Warning. Solving for level of Lake Superior without specifying net basin supplies. Null values have been assumed.
1123 Warning. Solving for level of Lake Superior without specifying Long Lac or Ogoki diversion data. Null values have been assumed.
1124 Tried to open a file for time-series output, but failed. The "Only One Open" flag is true, so this is unlikely to be a problem with too many files open. Another process (e.g., an editor) may already have it open, or there is a permission problem. The filename, IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), and the build information are as follows:
1125 Tried to open a file for time-series output, but failed. All output files are open simultaneously, so likely exceeded operating system's limit on open files. Test by including a parameter line such as:

Only One Open: True

Although "too many files open" is a common problem, this specific error could also be due to something else. Another process (e.g., an editor) may already have it open, or there may be a permission problem. The filename, IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), and the build information, are as follows:
1126 Tried to open a file for writing time-series output, but failed. File had been opened, written to, and closed successfully earlier in the run. Another process (e.g., an editor) may already have it open, or a disk error occurred.

The filename, IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), and the build information are as follows:
1127 Encountered write error. The IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line that caused the problem, are as follows:
1128 Requested an output file for a lake not being solved for. The lake and data requested are given below.
1129 Requested an improper output file. The period length (monthly, weekly, etc.) of the output file must be equal to or greater than the routing period used for the lake.
1130 Did not specify previous Lake Superior outflow, as required to use Plan 1977A. This is required information for the plan. Insert a line such as:

Prev. St. M. Riv. Flow: 1560 m3s

1131 Lake Ontario listed as one of the lakes to solve for, but failed to specify how to compute Lake Ontario outflows. Insert a USTLRIVSOLUTION record.
1132 Warning. Solving for level of Lake Ontario without specifying net basin supplies. Null values will be assumed.
1133 Warning. Solving for level of Lake Ontario without specifying Welland Canal flow data. Null values will be assumed.
1134 USTLRIVSOLUTION record specified an invalid value. This parameter tells how to compute Lake Ontario outflows. If the characters after the ':' are '1958D', then the program uses Plan 1958D to calculate outflows. If '1998', then Plan 1998 is used to calculate Ontario outflows. If an 'O', an outflow equation is used (that must be specified using the USTLRIVEQN parameter). The value read and the input line are as follows:
1135 Warning. P77AMONTHSTOAVG record failed to include a proper list of how many months of forecasted outflows must be averaged to compute a Plan 1977A outflow. The comma-delimited list should contain 12 positive integers. The first shows the number of months to average in order to compute a January outflow, the second shows the number of months to average a February outflow, etc. This is optional information for using Plan1977A; if not specified the program assumes 5 months to average, no matter which month is being computed. Program terminated. Please check syntax of input. The incorrect input line is as follows:
1136 A P77AMONTHSTOAVG record contains an improper value. Only possible to average between 1 and 12 months of Plan 1977A outflows to compute the final Plan 1977A flow. The incorrect input line is as follows:
1137 Input record with comma-delimited list of integer input values does not have enough values on the line.
1138 Could not read field in a comma-delimited list as an integer (I6).
1139 P77A MidLake Inc record specifies an invalid value. The number of increments must be between 1 and 999. The value read and the input line are as follows:
1140 Could not read value after the colon as an integer (I9).
1141 Computed value unrealistic. There are two possible explanations. A lake is being drained or filled up well beyond physical possibility. Check the input data (especially units).

Otherwise, the model is somehow using the missing data marker (-9999) as valid input. Though a bug is possible, this situation could happen if some column of input data (with mixed valid and missing data) needed conversion, either into a different routing interval, or different units. When specifying forced flows, always provide them in the base units (m or m3s), and in the proper routing interval.
1142 Tried to specify a time series of daily values as repeating data. This is not supported simply because the number of values is so cumbersome. Daily time-series values must be specified using an input data file. The end of the line in question looks like:
1143 Failed to parse a line of daily input data according to the standard format (I4,I3,I2,8F8). Simulation end date may be beyond data provided. Check also for garbled data. The IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line in question, are as follows:
1144 Failed to parse a line of daily input data according to the standard format (I4,I3,I2,8F8). The year, month, and quarter-month given on the line were not sequential from that of the previous line. Check that file is in daily format, contains no missing lines, is not in a different format, or was corrupted.
1145 Water-balance or retardation time-series of input data contains only missing values for a routing period. This will yield spurious results. Only 'special' columns of data may contain missing values. Printed below are the lake name, period ending date, and column number from the lake's summary input file:
1146 'Compatibility mode' for Plan 1977A is set to do a six-month regulation run, but the end date is set beyond six months. (There are only six months of NBS in the operational supply matrix).

1147 Only weekly routing through Lake Ontario is permitted if using Plan 1958D. Separating the routing period from the regulation period is not yet permitted for Lake Ontario.
1148 'P77A Compat Mode' record specifies an invalid value. The value must be an integer (i.e., specifically -1, 0, or 1). Read the documentation for option details. The value specified was:
1149 Plan 1958D is implemented only for weekly regulation, with the week beginning on a Friday. You must specify the year, month, and date of a Friday for the P58D Forecast Date. The line was given as:
1150 Could not find the Plan 1958D Simulation input file specified by parameter P58DSIMULATEFILE. The filename used and the parameter line it was read from are as follows:
1151 Specified a time series of weekly values as repeating data. This is not supported, because weeks are not repeatable on an annual basis. Weekly time-series values must be specified using an input data file. The end of the line in question looks like:
1152 Failed to parse a line of daily input data according to the standard format (I4,I3,I2,12F8). The year, month, and quarter given on the line were not sequential from that of the previous line, or contained illegitimate values. File may be in some other format, or has missing lines, or was corrupted.
1153 Failed to parse a line of weekly input data according to the standard format (I4,I3,I3,1F8). The year, month, and date given on the line were not sequential from that of the previous line, or contained illegitimate values. File may be in some other format, or has missing lines, or was corrupted.
1154 Failed to parse a line of weekly input data according to the standard format (I4,I3,I3,1F8). End of final routing period may be beyond data provided. Check also for garbled data. The IOSTAT value (which has meaning only to the FORTRAN compiler used to create this version of the CGLRRM executable), the build information, and the line in question, are as follow:
1155 Appears to be a discontinuity in the dates appear inside a weekly input data file. The dates expected and the date encountered are listed below:
1156 Water-balance or retardation time-series of input data contains some missing values for a routing period. The CGLRRM doesn't estimate data. Only 'special' columns of data may contain missing values. Printed below are the lake name, period ending date, and column number from the lake's summary input file:
1157 Failed to parse a line of quarter-monthly input data according to the standard format (I4,I2,12F8). The year and quarter given on the line were not sequential from that of the previous line. File may be in different format, have missing lines, or was corrupted.
1158 Inconsistent input detected regarding lake area data. An area record for a lake not being routed was supplied. Please review your input parameters and/or the detailed log for more information. The number of points and lake in question are:
1159 Failed top specify area data for a lake you are routing through. Provide a record such as

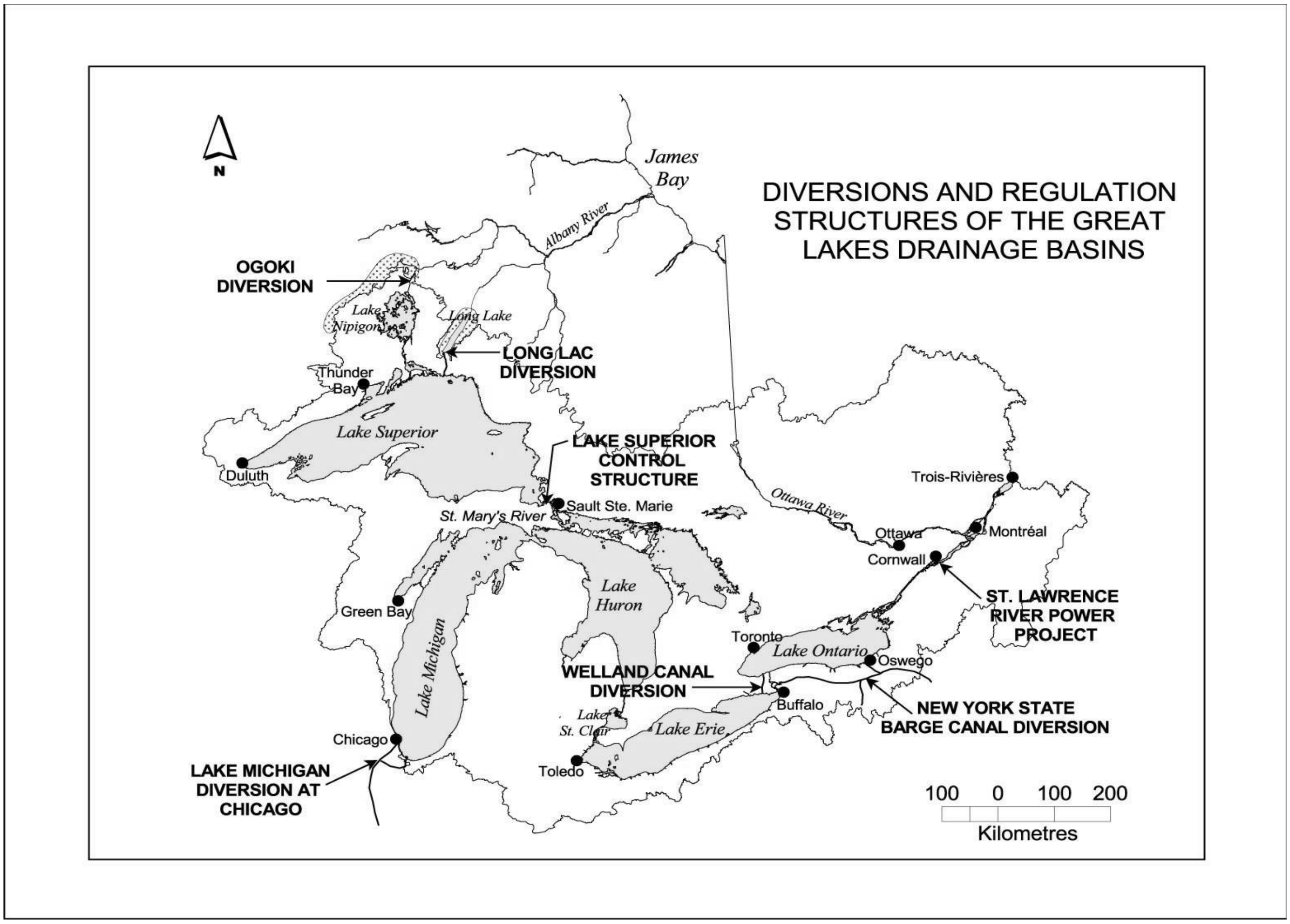
    Sup. Area/Elev: 82521.429480 km2
    MHu Area-elev: 117470753820 m2

This is only a sample. In your case, the missing area data is for
1160 St. Mary’s Solution parameter indicates using an outflow equation to determine St. Mary’s River flows, but either the equation was not specified, or wasn't specified correctly.
1161 Warning. Calculating outflows for a lake without specifying any retardation losses (i.e., for Lake Erie, generally would have at least one of "Nia Riv Ice Data", "Nia Riv Weed Data", or "Nia Riv Retr Data"). Null values have been assumed. This warning pertains to:
1162 Trying to route through the middle lakes, but didn't specify one of the outflow equations.
1163 Computing Lake Superior outflows using Plan 1977-A, but input contained a "St M Riv Eqn" parameter, specifying a discharge equation for the St. Mary’s River.
1164 Plan 1977A needs to know beginning-of-month levels. Since this run starts at the beginning of a month, it uses the initial lake levels for the run, and you don't need to explicitly provide BOM levels for Plan 1977A (i.e., "Sup BOM Level", "MHU BOM Level", etc.). However, you specified a value for at least one of them anyway, and it contradicts the initial lake level, causing ambiguity and resulting in the sad demise of your run.
1165 Tried to use an input parameter that is not fully implemented. It has been registered in the list of allowable parameters, but no code has yet been implemented to parse the value(s). Please inform the model developers of your desire.
1166 Missing information: don't have either the run's Start Date or End Date completely defined. Check your input.
1167 Error. Your input indicates not solving for at least one of the middle lakes. You must be routing through Lake Superior and the middle lakes in order to use Plan 1977A.
1168 Error. You must specify a value for parameter SUPCRITERIAROUND.
1169 Error. You must specify a value for parameter SUPSWPIERFORMULA.

1170 'SUPSWPIERFORMULA' record specifies an invalid value. The value must be an integer (specifically, 1 or 2). Use 1 for the old, traditional formula, and 2 for the new EC formula derived in 2000. The value specified was:
1171 The model is coming up with unreasonable numbers for Lake Superior, indicating a bust somewhere. This particular failure was triggered by a level too high to stay under the maximum outflow constraints in Plan 1977A. The related values are listed below. However, this is more likely a symptom than the cause. Review the results thus far to see when the lake levels began climbing out of a reasonable range, and try to determine why. Check units on input data, turn of the Sup Verbosity parameter, etc.
1172 'SUPCRITERIAROUND' record specifies an invalid value. The value must be an integer (specifically, 1 or 2). This parameter directs the program on to round calculations regarding Criterions B and C, and also while determining total non-gated flow. A value of 1 means to use the rounding practices embedded in Plan 1977A. A value of 2 means to use the system-wide settings established with the parameters 'LEVELROUND', etc). The value specified was:
1173 You tried to enter the same parameter twice. Only GENERALVERBOSITY, MODELERNAME, the parameters specifying area-elevation pairs, and the parameters regarding repeating time-series data are allowed multiple entries.
1174 You entered a parameter that the model accepts for input, but doesn't currently use for anything. This situation occurs because of a partial implementation of some anticipated feature, or because some change or disablement may not be permanent. The run stops so that you can verify your input. The parameter that caused the problem is shown below.
9999 10 20 30 40 50 60 70 80 90
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345678901234567890

Appendix D

MAP OF THE GREAT LAKES BASIN


DIVERSIONS AND REGULATION STRUCTURES OF THE GREAT LAKES DRAINAGE BASINS
N
James Bay
Albany River
OGOKI DIVERSION
Lake Nipigon
Long Lake
LONG LAC DIVERSION
Thunder Bay
Lake Superior
Duluth
LAKE SUPERIOR CONTROL STRUCTURE
Sault Ste. Marie
St. Mary's River
Ottawa River
Trois-Rivières
Ottawa
Montréal
Cornwall
ST. LAWRENCE RIVER POWER PROJECT
Green Bay
Lake Huron
Lake Michigan
Toronto
Lake Ontario
Oswego
WELLAND CANAL DIVERSION
Buffalo
NEW YORK STATE BARGE CANAL DIVERSION
Lake St. Clair
Lake Erie
Chicago
Toledo
LAKE MICHIGAN DIVERSION AT CHICAGO
100 0 100 200
Kilometres

Appendix E

DESIGN OBJECTIVES

Specific design objectives include the following:

- Thorough Documentation – The model required an informative user's manual describing the various options and requirements, with plenty of examples. The code, too, required full internal documentation. Dissatisfaction with undocumented codes of predecessor models was one of the reasons for undertaking the CGLRRM. This user's manual is a stand-alone document describing model organization and processing, computations, data formats, limitations, comprehensive input description, and recommended methods for applying the CGLRRM to various objectives, including detailed walk-through of several example problems.
- Robust Error Trapping and User Feedback - This critical objective has been met. The only times program failure results in a cryptic Fortran error message are related to problems with file input and output.
- Modularity – Required distinct components which could be modified individually, without modifications creating impacts in seemingly unrelated modules throughout the code (e.g., change Plan 1977-A (Lake Superior regulation) module or vary its input parameters, without affecting Lake Ontario code). This objective was achieved, but with tradeoffs regarding integration and performance.
- Fortran 77 – This programming language was used since it is commonly used.
- Portability – The model can be run on any MS-DOS or Windows PC, Unix workstations, or Macintosh systems, without significant modifications required. Use of compiler extensions beyond ANSI standard has been minimized. It was very important to get the same answer on different platforms, as improper 64-bit arithmetic in earlier models led to coordination problems. Since the CGLRRM code itself was coordinated, it was crucial to do it correctly. This does not apply to the GUI.
- Holistic Routing – It is possible to route from Lake Superior through Trois-Rivieres (or any particular segment in between) for each time step. However, there were tradeoffs with regard to modularity.
- Perform Both Simulation and Forecasting – The model is flexible enough to route long periods of given historical or future supplies and compute statistics, as well as accommodate the minimal input required for an operational run (where the time-series input data is implied by a regulation plan). Dissatisfaction and confusion with the multiple versions of the earlier models is one the main reasons the CGLRRM was undertaken. Achieving such flexibility resulted in increased model complexity, however.
- Complete Parameterization – Most of the hardcoded values are items such as metric conversions and seconds in a day. Most regulation and routing parameters must be read in (though some regulatory parameters are hardcoded to default values) . This was a significant flexibility objective that avoided recompiling and minimizes problems associated with studies that vary constants, etc. The compromise is that the extra complexity regarding input burdens the user. Thorough documentation and good implementation of separate/protected parameter files has minimized this concern. Since almost everything is input and nothing is presumed, the next objective becomes very important.
- Input Data Flexibility – It is possible to mix and match monthly, quarter-monthly, weekly, and daily times-series input data, regardless of the simulation interval, and without the data files being cropped to the routing period. Previous models had rigid input requirements resulting in multiple versions of the same data for different routing intervals, which created confusion. Again, achieving this flexibility resulted in increased complexity.

- Routing Period Flexibility – The user is able to select monthly, quarter-monthly, weekly, or daily routing interval for Lakes Superior, Michigan through Erie, or Ontario, as appropriate. Previous models spawned multiple versions for different routing intervals, causing confusion and hindering maintenance. Again, achieving this flexibility resulted in increased complexity.
- No Fixed Limit on Routing Intervals – The model can route thousands of years of supplies in one run. Providing this ability meant a change from the simple approach used in previous models, which was to store all inputs and results in memory simultaneously. Intermediate values would have to be written to disk, and re-read when needed later. Rather than get into dynamic disk-swapping memory management (given Fortran 77), and in view of the limitations regarding open files, the CGLRRM loops through the simulation period three times. The first loop is for inflow pre-processing. All the supplies, diversions, etc. for each lake are added up and summarized in one file per lake. (The ice/weed retardation values are summarized in the same file too.) This summary can be re-used between runs, if the supplies, etc., and start date are the same. The second pass reads the summarized supplies and performs the routing computations, writing the resulting water levels and flows to disk. The third pass calculates statistics and generates the custom reports. The tradeoffs to this feature are decreased performance and increased complexity.
- Optional Extremely Detailed Output – If desired, the user may trace each step of all computations.
- Repeatable or Explicit Time-Series Input – Users may either directly specify diversions, etc., for each routing period, or use some scheme for repeating them, such as providing 12 monthly values to use for every year (Plan 1977-A inputs insist on this ability). This objective has been achieved for monthly and quarter-monthly data, but not weekly or daily values yet.
- Simplicity - A fundamental law of technology states that complexity is proportional to capability. This objective was therefore often overridden by demand for some extra ability. However, as the CGLRRM grew larger and more complicated, the complexity was weighed objectively, and had to be warranted. Though, simplicity is largely subjective, the model was kept as simple as possible in this manner.
- Input Mirrored in Output - Detailed output files include a summary of what inputs and parameters were used. The only tradeoff was that it took much longer to code in this feature.
- Use Pre-Project Equations or Regulation Plans – The model is modular enough to compute Lake Superior outflows using the 1887 equation or Plan 1977-A.
- Lake Erie Regulation Module – Again the model modularity enables this option to be added in the future, if desired, with minimal coding.
- Fast – Model run speeds are currently very good. However, extra features will impact this, though most of today's computers will not be affected.
- Log and Report which Plan 1977-A Criteria Affect Decision. This objective has been met.
- Force Outflow or Gate Positions – The outflow objective has been achieved and the gate position forcing option will be implemented soon.
- Detail St. Mary's Outflow Split – This objective has yet to be addressed
- Compute Plan 1977-A Side Channel Flows Based on Pool Levels – Here, the procedure has been enabled, but not coded so far.

- Computations Account for Variations in Lake Surface Area – This option has been achieved.
- Strict Data File Format Conventions - One format is used for each data interval for both input and output files.
- All Data Files ASCII – This objective has been achieved.
- Accept Component Supplies – This option has been achieved.
- Mandatory Specification of Units – This requirement has been achieved.
- Strict Data File Naming Conventions – This requirement has been achieved.
- Perform Probabilistic Analyses – This option is too complex to internalize in Fortran code. Currently, using macros to loop the routing model, and computing statistics externally is required.
- Attractive, Customized Output – Currently, this is only possible if done outside the CGLRRM.

Appendix F

ALLOWABLE FILENAMES FOR TIME-SERIES DATA

Below is a comprehensive list for all time-series data (input and output). The first two characters denote the lake or river for which the data is provided. Filenames have an "m", "q", "w", or "d" as the third character (although only the quarter-monthly "q" version of the filename is listed in Appendix F). The seventh and eighth positions, and the three-character extension, are available for optional descriptions. The "?" symbols in the names below act as placeholders.

spqtsd??.??? Lake Superior summary file of time-series input data
spqnbs??.??? Lake Superior net basin supplies
spqmlv??.??? Lake Superior water levels (period average)
spqelv??.??? Lake Superior water levels (end of period)
spqcon??.??? Lake Superior consumptive use
spqllk??.??? Lake Superior diversions from Long Lake
spqogo??.??? Lake Superior diversions from Ogoki
spqllo??.??? Combined Lake Superior diversion from Long Lake and Ogoki
spqdiv??.??? Lake Superior miscellaneous diversions (other than Long Lake and Ogoki)
spqdev??.??? Lake Superior deviations
spqfof??.??? Lake Superior forced outflows
spqfgo??.??? Lake Superior forced gates open
spqfsc??.??? Lake Superior forced sidechannel flows
spqgdw??.??? Lake Superior net groundwater contribution
spqrun??.??? Lake Superior runoff from tributaries
spqpre??.??? Lake Superior precipitation
spqevp??.??? Lake Superior evaporation
spqdev??.??? Lake Superior deviations from regulation plan

mrqflo??.??? St. Mary's River flow
mrqice??.??? St. Mary's River ice retardation (flow)
mrqwee??.??? St. Mary's River weed retardation (flow)
mrqciw??.??? St. Mary's River combined weed and ice retardation (flow)
mrqglp??.??? St. Mary's River flow through Great Lakes Power Limited (Canada)
mrqusp??.??? St. Mary's River flow through U.S. government plant (including Unit 10)
mrqesp??.??? St. Mary's River flow through Edison Sault Electric Company (U.S.)
mrqthp??.??? Total St. Mary's River hydropower flow
mrqcon??.??? St. Mary's River flow used by cities and industry
mrqrpd??.??? St. Mary's River flow through the rapids
mrqfis??.??? St. Mary's River flow through fisheries works
mrqtcw??.??? Total St. Mary's River flow through Compensating Works
mrqcnl??.??? St. Mary's River flow through Canadian lock
mrqusl??.??? St. Mary's River flow through U.S. locks

mhqtsd??.??? Lakes Michigan-Huron summary file of time-series input data
mhqnbs??.??? Lakes Michigan-Huron net basin supplies
mhqchi??.??? Lakes Michigan-Huron diversion at Chicago
mhqcon??.??? Lakes Michigan-Huron consumptive use
mhqdiv??.??? Lakes Michigan-Huron other diversions
mhqsup??.??? Lakes Michigan-Huron other supply
mhqmlv??.??? Lakes Michigan-Huron water levels (period average)

mhqelv??.??? Lakes Michigan-Huron water levels (end of period)
mhqgdw??.???Lakes Michigan-Huron net groundwater contribution
mhqrun??.??? Lakes Michigan-Huron runoff from tributaries
mhqpre??.??? Lakes Michigan-Huron precipitation
mhqevp??.??? Lakes Michigan-Huron evaporation

miqnbs??.??? Lake Michigan net basin supplies
miqcon??.??? Lake Michigan consumptive use
miqdiv??.??? Lake Michigan other diversions
miqsup??.??? Lake Michigan other supply
miqgdw??.??? Lake Michigan net groundwater contribution
miqrun??.??? Lake Michigan runoff from tributaries
miqpre??.??? Lake Michigan precipitation
miqevp??.??? Lake Michigan evaporation

huqnbs??.??? Lake Huron net basin supplies
huqcon??.??? Lake Huron consumptive use
huqdiv??.??? Lake Huron other diversions
huqsup??.??? Lake Huron other supply
huqgdw??.??? Lake Huron net groundwater contribution
huqrun??.??? Lake Huron runoff from tributaries
huqpre??.??? Lake Huron precipitation
huqevp??.??? Lake Huron evaporation

crqflo??.??? St. Clair River flow
crqice??.??? St. Clair River ice retardation (flow)
crqwee??.??? St. Clair River weed retardation (flow)
crqciw??.??? St. Clair River combined weed and ice retardation (flow)

scqtsd??.??? Lake St. Clair summary file of time-series input data
scqnbs??.??? Lake St. Clair net basin supplies
scqcon??.??? Lake St. Clair consumptive use
scqdiv??.??? Lake St. Clair miscellaneous diversions
scqsup??.??? Lake St. Clair other supplies
scqmlv??.??? Lake St. Clair water levels (period average)
scqelv??.??? Lake St. Clair water levels (end of period)
scqgdw??.??? Lake St. Clair net groundwater contribution
scqrun??.??? Lake St. Clair runoff from tributaries
scqpre??.??? Lake St. Clair precipitation
scqevp??.??? Lake St. Clair evaporation

drqflo??.??? Detroit River flow
drqice??.??? Detroit River ice retardation (flow)
drqwee??.??? Detroit River weed retardation (flow)
drqciw??.??? Detroit River combined weed and ice retardation (flow)

erqtsd??.??? Lake Erie summary file of time-series input data
erqnbs??.??? Lake Erie net basin supplies
erqwel??.??? Lake Erie diversion through Welland Canal
erqdiv??.??? Lake Erie other diversions
erqsup??.??? Lake Erie other supplies
erqcon??.??? Lake Erie consumptive use
erqmlv??.??? Lake Erie water levels (period average)
erqelv??.??? Lake Erie water levels (end of period)
erqgdw??.??? Lake Erie net groundwater contribution
erqrun??.??? Lake Erie runoff from tributaries
erqpre??.??? Lake Erie precipitation
erqevp??.??? Lake Erie evaporation

nrqflo??.??? Niagara River flow
nrqice??.??? Niagara River ice retardation (flow)
nrqwee??.??? Niagara River weed retardation (flow)
nrqciw??.??? Niagara River combined weed and ice retardation (flow)
nrqbrl??.??? Niagara River flow through Black Rock Lock

onqtsd??.??? Lake Ontario summary file of time-series input data
onqnbs??.??? Lake Ontario net basin supplies
onqmlv??.??? Lake Ontario water levels (period average)
onqelv??.??? Lake Ontario water levels (end of period)
onqcon??.??? Lake Ontario consumptive use
onqdiv??.??? Lake Ontario miscellaneous diversions
onqgdw??.??? Lake Ontario net groundwater contribution
onqrun??.??? Lake Ontario runoff from tributaries
onqpre??.??? Lake Ontario precipitation
onqevp??.??? Lake Ontario evaporation

There are a number of inputs defined for the St. Lawrence River that will be added to this list in the near future.